

Advanced Machine Learning

Entropy concepts used for machine learning

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2. Entropy measures for grammatical models
3. Maximum entropy models
4. Regularized EM algorithm
5. Discriminative training criterion
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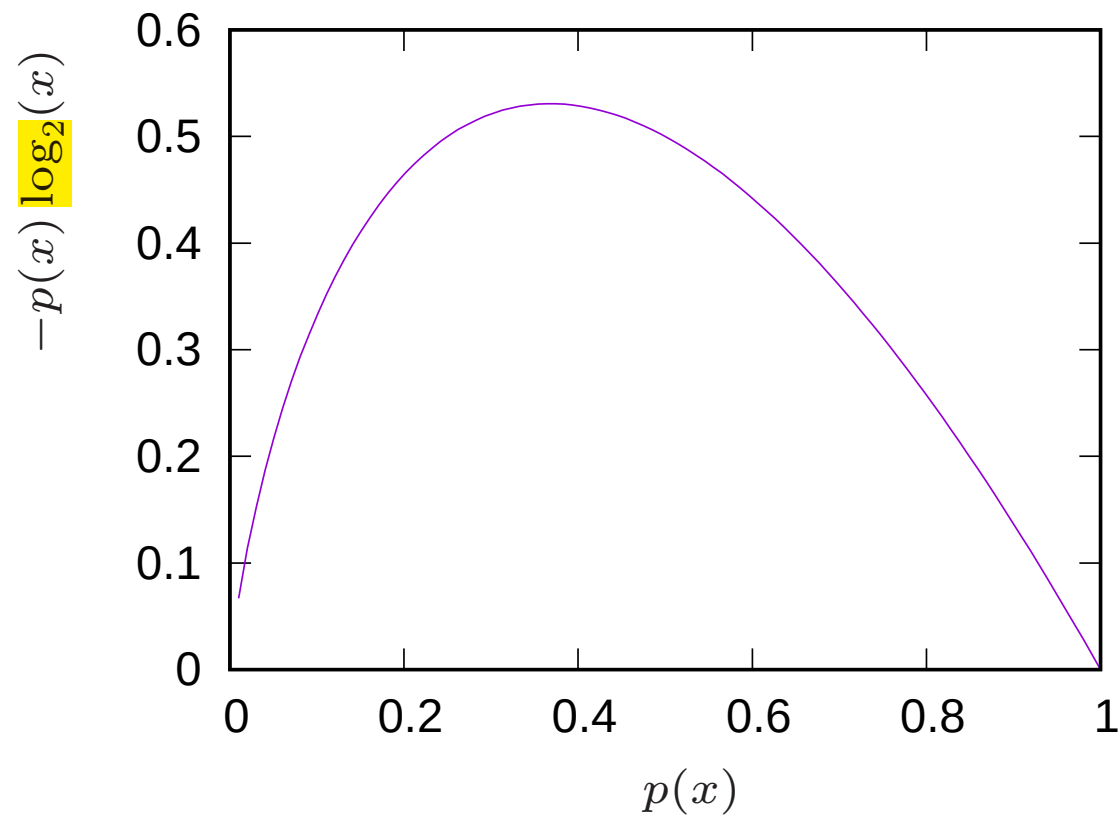
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Entropy definitions

Entropy of a discrete variable:

$$H(X) = - \sum_x p(x) \log p(x) \quad (1)$$



Entropy definitions

If the log is to the base 2 then the entropy is measured in bits

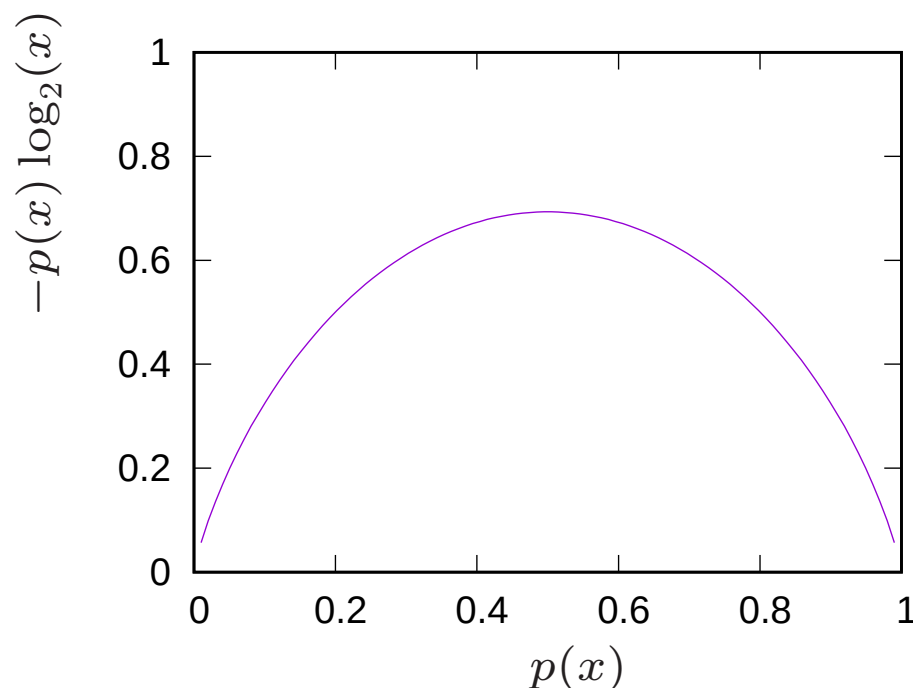
Interpretation 1: average number of bits needed to describe X

Interpretation 2: *uncertainty* about the outcome of a stochastic variable X

Property: entropy is a concave function

Property: entropy is maximum when p is uniform $\rightarrow$ maximum *uncertainty*

Example: a coin with 2 faces and identical probability



Entropy definitions

Example [Cover 1991]: Consider a dice with 8 faces (with numbers from 0 to 7), all of them with the same probability. A 3-bit string is necessary as label for each face. The entropy of this random variable is:

$$H(X) = - \sum_{i=1}^8 p(i) \log p(i) = - \sum_{i=1}^8 \frac{1}{8} \log \frac{1}{8} = \log 8 = 3 \text{ bits}$$

Intuition: let us suppose that we send the winner face to a friend. The average number of bits we need to send to the friend to identify the face is 3 bits.

Let us suppose that:

$$P(X = 0) = \frac{1}{2}, \quad P(X = 1) = \frac{1}{4}, \quad P(X = 2) = \frac{1}{8}, \quad P(X = 3) = \frac{1}{16}$$

$$P(X = 4) = P(X = 5) = P(X = 6) = P(X = 7) = \frac{1}{64}$$

Then, the entropy is: $H(X) = 2 \text{ bits}$

Intuition: the most probable outcomes need less bits to be coded. Consider the following coding for the faces: {0, 10, 110, 1110, 111100, 111101, 111110, 111111}

$$\frac{1}{2} + 2 \frac{1}{4} + 3 \frac{1}{8} + 4 \frac{1}{16} + 4 \cdot 6 \frac{1}{64} = 2$$

Entropy definitions

Joint entropy of a pair of discrete random variables (X, Y) with a joint distribution $p(x, y)$ [Cover 1991]:

$$H(X, Y) = - \sum_x \sum_y p(x, y) \log p(x, y) \quad (2)$$

Conditional entropy of a pair of discrete random variables (X, Y) :

$$H(Y|X) = \sum_x p(x) H(Y|X = x) \quad (3)$$

$$= - \sum_x p(x) \sum_y p(y|x) \log p(y|x) \quad (4)$$

$$= - \sum_x \sum_y p(x, y) \log p(y|x) \quad (5)$$

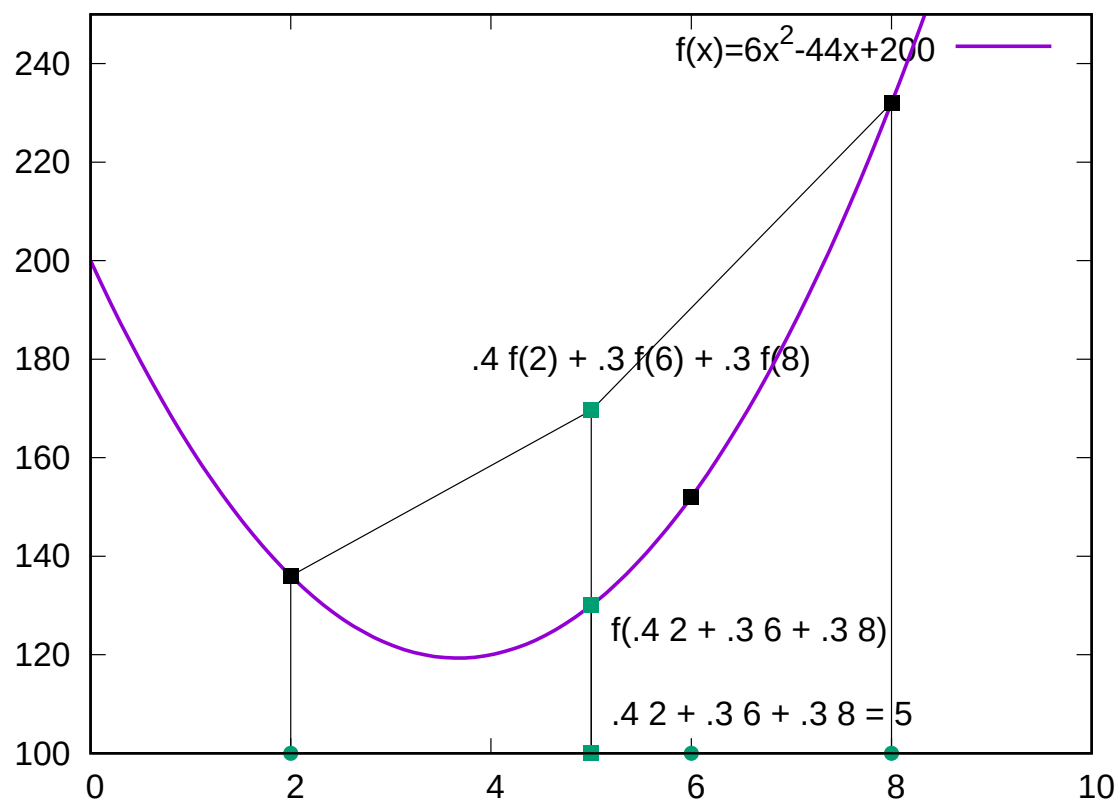
Theorem [Cover 1991]

$$H(X, Y) = H(X) + H(Y|X) \quad (6)$$

Entropy definitions

Theorem. (Jensen's inequality) Let f be a convex function defined on an interval I . If $x_1, x_2, \dots, x_n \in I$ and $\lambda_1, \lambda_2, \dots, \lambda_n \geq 0$ with $\sum_{i=1}^n \lambda_i = 1$, then,

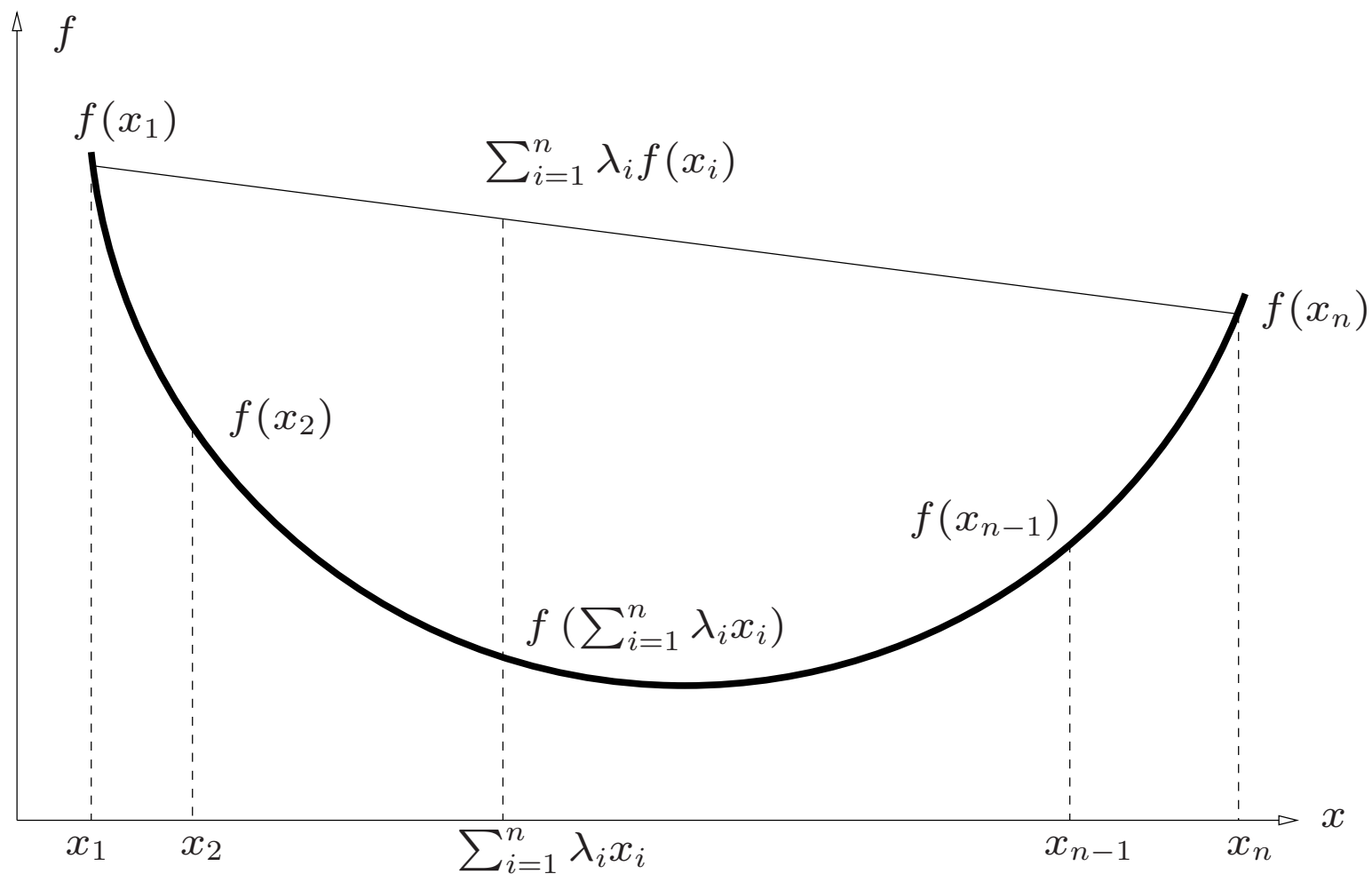
$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i). \quad (7)$$



$$\lambda_1 = .4 \quad \lambda_2 = .3 \quad \lambda_3 = .3$$

Entropy definitions

Graphical interpretation of Jensen's inequality



Entropy definitions

Relative entropy or *Kullback-Leibler distance* between two probability mass functions $p(x)$ and $q(x)$:

$$D(p||q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \quad (8)$$

- Non-negative

$$\begin{aligned} -D(p||q) &= -\sum_x p(x) \log \frac{p(x)}{q(x)} = \sum_x p(x) \log \frac{q(x)}{p(x)} \leq \log \sum_x p(x) \frac{q(x)}{p(x)} \\ &= \log \sum_x q(x) = \log 1 = 0 \end{aligned}$$

The inequality follows from Jensen's inequality

- Non-symmetric and does not satisfy the triangle inequality

Exercise ():** Demonstrate if $D(p||q) + D(q||p)$ is or is not symmetric and satisfy the triangle inequality.

Entropy definitions

Mutual information between a pair of discrete random variables (X, Y) :

$$I(X; Y) = \sum_x \sum_y p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \quad (9)$$

Interpretation: It measures how much knowing one of these variables reduces uncertainty about the other¹

$$I(X; Y) = H(X) - H(X|Y) \quad (10)$$

$$= H(Y) - H(Y|X) \quad (11)$$

Property:

$$I(X; Y) = D(p(x, y) || p(x)p(y)) \geq 0$$

¹From [wikipedia](#): For example, if X and Y are independent, then knowing X does not give any information about Y and vice versa, so their mutual information is zero. At the other extreme, if X is a deterministic function of Y and Y is a deterministic function of X then all information conveyed by X is shared with Y : knowing X determines the value of Y and vice versa. As a result, in this case the mutual information is the same as the uncertainty contained in Y (or X) alone, namely the entropy of Y (or X)

Entropy definitions

Interpretation of mutual information in special cases

- X and Y independent:

$$\begin{aligned}
 I(X; Y) &= H(Y) - H(Y|X) = H(Y) - \sum_x p(x) H(Y|X = x) \\
 &= H(Y) + \sum_x p(x) \sum_y p(y|x) \log p(y|x) = H(Y) + \sum_x p(x) \sum_y p(y) \log p(y) \\
 &= H(Y) - H(Y) \sum_x p(x) = 0
 \end{aligned}$$

- X and Y are “totally dependent” (or the same variable, see comments above):

$$\begin{aligned}
 I(X; X) &= H(X) - H(X|X) = H(X) - \sum_x p(x) H(X|X = x) \\
 &= H(X) - \sum_x p(x) \sum_x p(x|x) \log \underbrace{p(x|x)}_{=1^{(*)}} = H(X)
 \end{aligned}$$

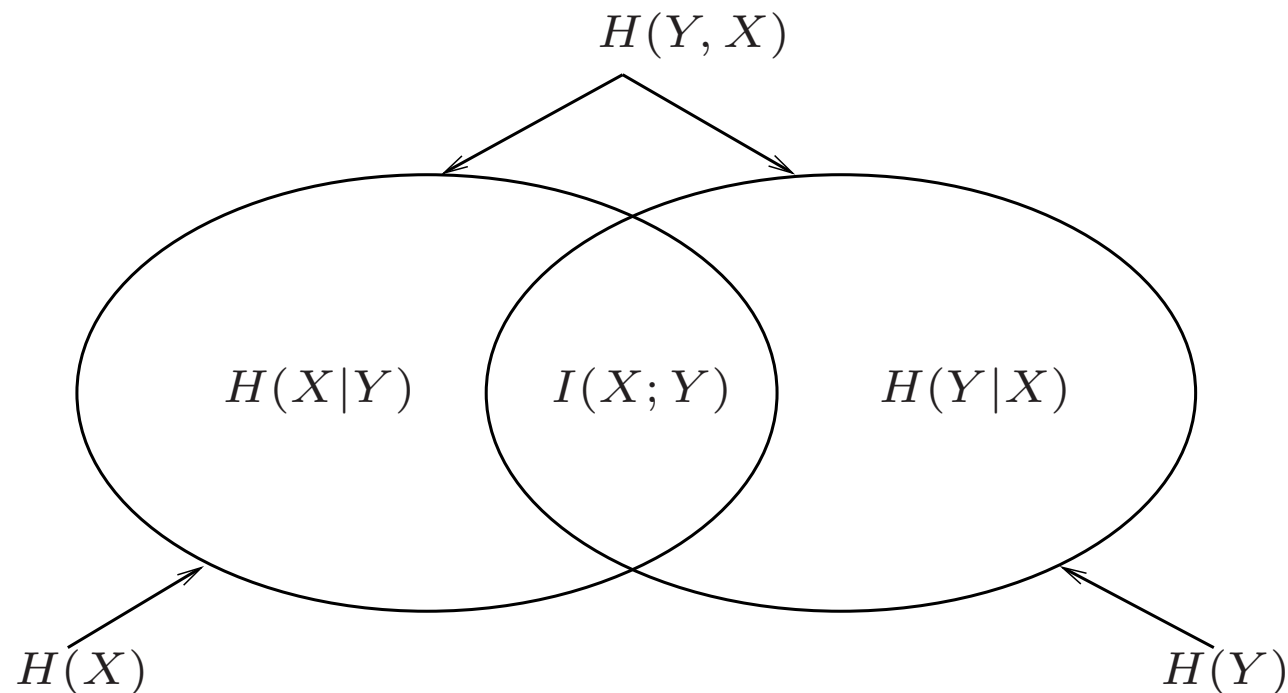
$(*) P(A|B) = \frac{P(A \cap B)}{P(B)} \underset{A=B}{=} \frac{P(B)}{P(B)} = 1$

Entropy definitions

$$I(X; Y) = H(X) - H(X|Y) \quad (12)$$

$$= H(Y) - H(Y|X) \quad (13)$$

$$= H(X) + H(Y) - H(Y, X) \quad (14)$$



Entropy definitions

Example [Cover 1991]: Let (X, Y) have the following joint distribution:

$Y \setminus X$	1	2	3	4	Σ
1	1/8	1/16	1/32	1/32	1/4
2	1/16	1/8	1/32	1/32	1/4
3	1/16	1/16	1/16	1/16	1/4
4	1/4	0	0	0	1/4
Σ	1/2	1/4	1/8	1/8	1

Marginal of X : $(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$

Marginal of Y : $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

$$\begin{aligned}
 H(X|Y) &= \sum_{i=1}^4 p(Y = i) H(X|Y = i) \\
 &= \frac{1}{4}H\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right) + \frac{1}{4}H\left(\frac{1}{4}, \frac{1}{2}, \frac{1}{8}, \frac{1}{8}\right) + \frac{1}{4}H\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right) + \frac{1}{4}H(1, 0, 0, 0) \\
 &= \frac{11}{8} = 1.375 \text{ bits} \\
 I(X; Y) &= 0.375 \text{ bits}
 \end{aligned}$$

Entropy definitions

Interesting properties [Cover 1991]

- $I(X; Y) \geq 0$
- $H(X|Y) \leq H(X)$

Exercise (*): For a given value of $Y = y$, is it possible that $H(X|Y = y) \geq H(X)$? Provide a demonstration.

Exercise (*): Write an example similar to the example in the previous slide but changing the joint distributions, where variables have a real meaning (real sense) (e.g. alcohol in blood versus car accident, weather versus car accident, ...). Then compute the $H(X)$, $H(X, Y)$, $H(X|Y)$ and $I(X; Y)$.

Entropy definitions

Exercise (taken from SIN-ETSINF)

Random variables:

- *Climatología* (C): despejado ($_{DES}$), nublado ($_{NUB}$), lluvia ($_{LLU}$)
- *Luminosidad* (L): día ($_{DIA}$), noche ($_{NOC}$)
- *Seguridad* (S): desplazamiento seguro ($_{SEG}$), o con accidente ($_{ACC}$)

$P(c, s)$	DES	NUB	LLU
SEG	0.43	0.30	0.13
ACC	0.03	0.03	0.08
$\Sigma = 1$			

$P(l, s)$	DIA	NOC
SEG	0.57	0.29
ACC	0.05	0.09
		$\Sigma=1$

$P(c, l)$	DES	NUB	LLU
DIA	0.31	0.21	0.10
NOC	0.15	0.12	0.11
$\Sigma = 1$			

$P(s, c, l)$	DIA			NOC		
	DES	NUB	LLU	DES	NUB	LLU
SEG	0.30	0.20	0.07	0.13	0.10	0.06
ACC	0.01	0.01	0.03	0.02	0.02	0.05
$\Sigma = 1$						

Entropy definitions

Exercise (*): (choose only one of them)

Compute $H(C)$, $H(S)$, $H(C|S)$, $H(S|C)$, $H(C, S)$, and $I(C; S)$

Compute $H(L)$, $H(S)$, $H(L|S)$, $H(S|L)$, $H(L, S)$, and $I(L; S)$

Compute $H(L)$, $H(C)$, $H(L|C)$, $H(C|L)$, $H(L, C)$, and $I(C; L)$

Exercise ():** (choose only one of them)

Compute $H(C|L, S)$, $H(L|C, S)$, $H(S|C, L)$

Compute $H(C, L|S)$, $H(L, S|C)$, $H(S, C|L)$

Compute $I(C; L; S)$

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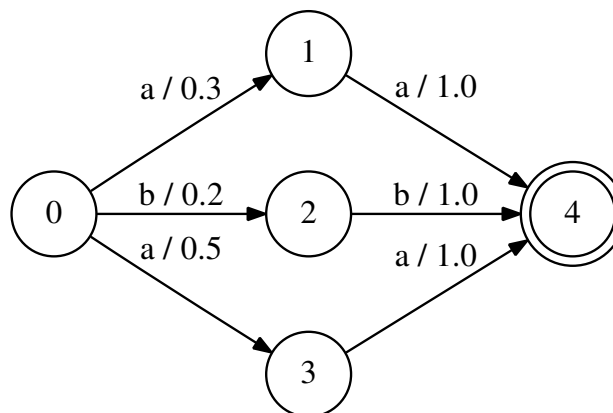
Entropy measures for grammatical models

Definition A *PFA* is a tuple $\mathcal{A} = \langle Q, \Sigma, \delta, I, F, P \rangle$, where: Q is a finite set of states; Σ is the alphabet; $\delta \subseteq Q \times \Sigma \times Q$ is a set of transitions; $I : Q \rightarrow \mathbb{R}^{\geq 0}$ is the probability function of a state being an initial state; $P : \delta \rightarrow \mathbb{R}^{\geq 0}$ is a probability function of probabilities transition between states; $F : Q \rightarrow \mathbb{R}^{\geq 0}$ is the probability function of a state being a final state.

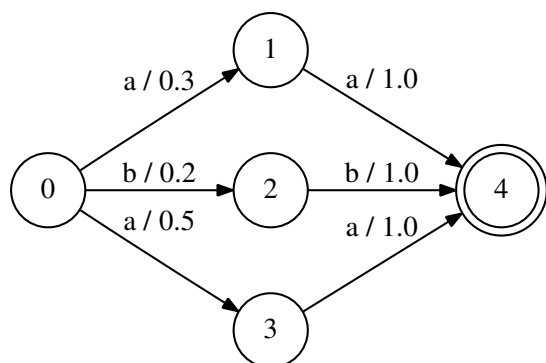
I , P , and F are functions such that:

$$\sum_{i \in Q} I(i) = 1 \quad (15)$$

$$\forall i \in Q, F(i) + \sum_{v \in \Sigma, j \in Q} P(i, v, j) = 1. \quad (16)$$



Entropy measures for grammatical models



- Sequence entropy [Grenander 1967]

$$H(\mathcal{A}) = - \sum_{w \in L(\mathcal{A})} p_{\mathcal{A}}(w) \log p_{\mathcal{A}}(w) \quad (17)$$

$$\begin{aligned} H(\mathcal{A}) &= -p_{\mathcal{A}}(aa) \log p_{\mathcal{A}}(aa) - p_{\mathcal{A}}(bb) \log p_{\mathcal{A}}(bb) \\ &= -0.8 \log 0.8 - 0.2 \log 0.2 = 0.72 \text{ bits} \end{aligned}$$

- Entropy on the paths given an observation sequence aa [Hernando 2005]

$$H_{\mathcal{A}}(\theta|w) = - \sum_{\theta \in \Theta_{\mathcal{A}}(w)} p_{\mathcal{A}}(\theta|w) \log p_{\mathcal{A}}(\theta|w) \quad (18)$$

$$\begin{aligned} H_{\mathcal{A}}(\theta|aa) &= -p_{\mathcal{A}}(034|aa) \log p_{\mathcal{A}}(034|aa) - p_{\mathcal{A}}(014|aa) \log p_{\mathcal{A}}(014|aa) \\ &= -\frac{0.5}{0.8} \log \frac{0.5}{0.8} - \frac{0.3}{0.8} \log \frac{0.3}{0.8} = 0.95 \text{ bits} \end{aligned}$$

- Derivational entropy [Grenander 1967]

$$H_d(\mathcal{A}) = - \sum_{\theta \in \Theta(\mathcal{A})} p_{\mathcal{A}}(\theta) \log p_{\mathcal{A}}(\theta) \quad (19)$$

$$\begin{aligned} H_d(\mathcal{A}) &= -p_{\mathcal{A}}(034) \log p_{\mathcal{A}}(034) - p_{\mathcal{A}}(014) \log p_{\mathcal{A}}(014) - p_{\mathcal{A}}(024) \log p_{\mathcal{A}}(024) \\ &= -0.5 \log 0.5 - 0.3 \log 0.3 - 0.2 \log 0.2 = 1.49 \text{ bits} \end{aligned}$$

Entropy measures for grammatical models

Computation of the derivational entropy

Given a *PFA* $\mathcal{A}$, we define the characteristic matrix M [Thompson 74] of dimensions $|Q| \times |Q|$ as²:

$$M(i, j) = \sum_{v \in \Sigma} P(i, v, j) . \quad (20)$$

The following vector ξ , $0 \leq i < |Q| - 1$ is defined [Soule 1974]:

$$\xi(i) = - \sum_{\substack{v \in \Sigma \\ 0 \leq j \leq |Q| - 1}} P(i, v, j) \log P(i, v, j) \quad (21)$$

and $\xi(|Q| - 1) = 0$.

Theorem [Grenander 1967] If the largest eigenvalue of the characteristic matrix M is strictly less than 1, then the derivational entropy of the model can be computed as: $H_d(\mathcal{A}) = (1 - M)^{-1} \xi$.

² Let's suppose that states are named with integers and there is only one initial state, 0, and one final state, $|Q| - 1$

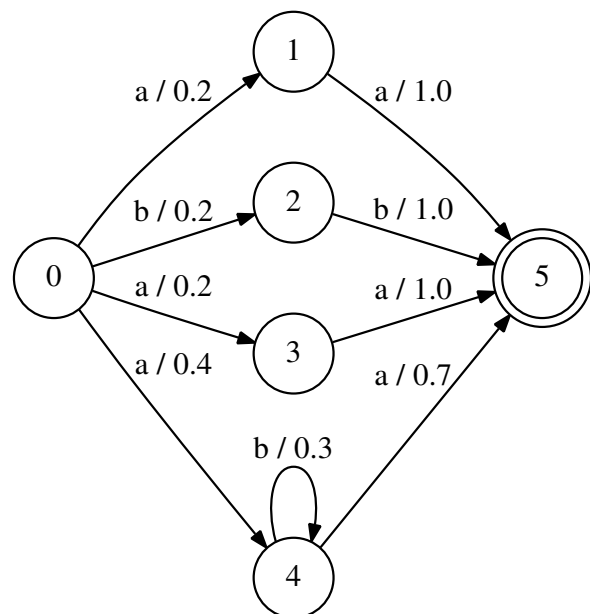
Entropy measures for grammatical models

Comments and additional properties:

- $H(\mathcal{A}) \leq H_d(\mathcal{A})$ with equality if the automaton is non-ambiguous [Grenander 1967]
- Computation of $H(\mathcal{A})$ is not feasible [Grenander 1967] in general (see example below)
- Computation of $H_d(\mathcal{A})$ is feasible in general with time complexity $O(|Q|^3)$ [Grenander 1967]. Less time complexity if the automaton is acyclic [Sánchez 2018]
- Computation of $H_{\mathcal{A}}(\theta|w)$ requires time complexity $O(|w||Q|^2)$ [Hernando 2005]

Entropy measures for grammatical models

A particular case for the computation of the sequence entropy $H(\mathcal{A})^1$



$$aa \longrightarrow -0.68 \log 0.68 = .3783$$

$$bb \longrightarrow -0.2 \log 0.2 = .4644$$

$$ab^n a \longrightarrow - \sum_{n>0} p(ab^n a) \log p(ab^n a) = 1.78$$

Exercise ():** Provide a much more complex example that has to include loops, cycles, ...

¹See Appendix 1 for some details.

Entropy measures for grammatical models

Exercise (**):** Adapt the computation of the entropy of the paths for a given sequence for PFA and apply the developed algorithm to a the given example. Take the following definitions as starting point and follow [Hernando 2005].

$$H_{\mathcal{A}}(\theta|w) = - \sum_{\theta \in \Theta_{\mathcal{A}}(w)} p_{\mathcal{A}}(\theta|w) \log p_{\mathcal{A}}(\theta|w)$$

Definitions:

- $H_t(j) = H(\theta_{1,t-1}|\theta_t = j, w_{1,t})$: entropy of all paths (state sequences) that lead to state j at time t , given the observation up to time t .
- $c_t(j) = p_{\mathcal{A}}(\theta_t = j|w_{1,t})$: probability of being in state j at time t , given the word sequence up to time t .

$$c_t(j) = \frac{\sum_{i=0}^{|\mathcal{Q}|-1} c_{t-1}(i) P(i, w_t, j)}{\sum_{k=0}^{|\mathcal{Q}|-1} \sum_{i=0}^{|\mathcal{Q}|-1} c_{t-1}(i) P(i, w_t, k)} \quad (22)$$

Auxiliary computations (see (6) in [Hernando 2005] for details):

$$p(\theta_{t-1} = i|\theta_t = j, w_{1,t}) = \frac{P(i, w_t, j) c_{t-1}(i)}{\sum_{k=0}^{|\mathcal{Q}|-1} P(k, w_t, j) c_{t-1}(k)} \quad (23)$$

Exercise ():** Demonstrate (23) for PFA.

Entropy measures for grammatical models

Initialization. For $0 \leq j < |\mathcal{Q}|$

$$H_0(j) = 0 \quad (24)$$

$$c_0(j) = I(j) \quad (25)$$

Recursion. For $0 \leq j < |\mathcal{Q}| - 1; 1 \leq t \leq |w|$:

$$c_t(j) = \frac{\sum_{i=0}^{|\mathcal{Q}|-1} c_{t-1}(i) P(i, w_t, j)}{\sum_{k=0}^{|\mathcal{Q}|-1} \sum_{i=0}^{|\mathcal{Q}|-1} c_{t-1}(i) P(i, w_t, k)}$$

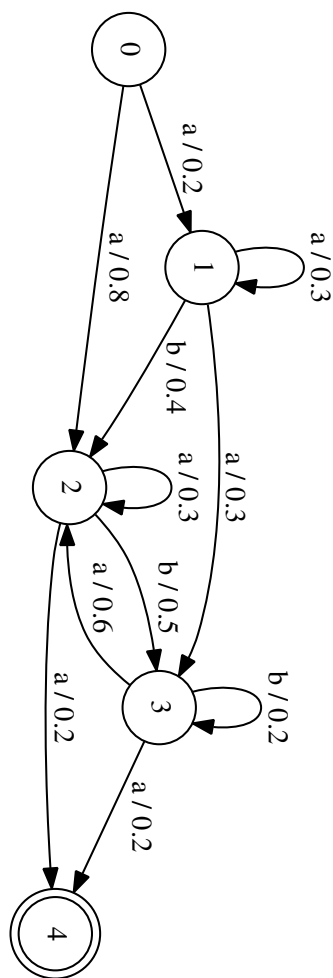
$$p(\theta_{t-1} = i | \theta_t = j, w_{1,t}) = \frac{c_{t-1}(i) P(i, w_t, j)}{\sum_{k=0}^{|\mathcal{Q}|-1} c_{t-1}(k) P(k, w_t, j)}$$

$$H_t(j) = \sum_{i=0}^{|\mathcal{Q}|-1} H_{t-1}(i) p(\theta_{t-1} = i | \theta_t = j, w_{1,t})$$

$$- \sum_{i=0}^{|\mathcal{Q}|-1} p(\theta_{t-1} = i | \theta_t = j, w_{1,t}) \log p(\theta_{t-1} = i | \theta_t = j, w_{1,t}) \quad (26)$$

Termination (to be completed).

Entropy measures for grammatical models



	a	a	b	a
0.0				
1.0				
0.0				
0.0	.2	.06/.52=.115		
0.0				.8143
0.0	.8	.24/.52=.462	.153	.735
0.0			.438	
0.0		.06/.52=.115	.847	
0.0				.987
0.0		.16/.52=.308		.265

 $H_t(0)$ $c_t(0)$ $H_t(1)$ $c_t(1)$ $H_t(2)$ $c_t(2)$ $H_t(3)$ $c_t(3)$ $H_t(4)$ $c_t(4)$

Entropy measures for grammatical models

Exercise (*): Define a PFA similar to the one that is in slide 15 and compute expressions (17)-(19). Choose a string with more than two derivations for (17).

Exercise (*): Compute the exact relation between the values in blue that are below expressions (17)-(19).

Exercise ():** Read [Hernando 2005]. Then, define a discrete HMM and compute $H_{\mathcal{A}}(\theta|w)$. The example, both the model and the string must be different from the example in [Hernando 2005].

Entropy measures for grammatical models

Exercise ():** (Practical exercise) Compute an approximation to expression (17) for an acyclic automaton $\mathcal{A}$ as follows:

1. Compute a list of n-best hypotheses ($\mathcal{B}$) for the automaton $\mathcal{A}$.
2. Compute $H_d(\mathcal{A})$ ($= H(\Theta)$) (19) (see [Sánchez 2018] for details).
3. Approximate $H(\Theta|\mathcal{A}) \approx -\sum_{w \in \mathcal{B}} p_{\mathcal{A}}(w) H_{\mathcal{A}}(\theta|w)$ with the n-best hypotheses of step 1.
4. Approximate expression (17) as follows:

$$\begin{aligned}
 H(\Theta|\mathcal{A}) + H(\mathcal{A}) &= H(\mathcal{A}|\Theta) + H(\Theta) \\
 H(\Theta|\mathcal{A}) + H(\mathcal{A}) &= \sum_{\theta \in \Theta(\mathcal{A})} p_{\mathcal{A}}(\theta) H_{\mathcal{A}}(\mathcal{A}|\theta) + H(\Theta) \\
 H(\Theta|\mathcal{A}) + H(\mathcal{A}) &= \sum_{\theta \in \Theta(\mathcal{A})} p_{\mathcal{A}}(\theta) \sum_{w \in L(\mathcal{A})} p_{\mathcal{A}}(w|\theta) \underbrace{\log p_{\mathcal{A}}(w|\theta)}_{\approx 0} + H(\Theta) \\
 H(\mathcal{A}) &= H(\Theta) - H(\Theta|\mathcal{A})
 \end{aligned} \tag{27}$$

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Maximum entropy models

- **Problem:** estimate $p(y|x)$ where $x = (x_1, \dots, x_D)$ is D -tuple of discrete (some times, categoric) observations

$$p(y|x_1, \dots, x_D)$$

- **Maximum entropy solution:** log-linear combination
- **Pending problems:**
 - Selection of features
 - Smoothing: no closed-form solutions for optimal parameters

Maximum entropy models

Conditional entropy (see (4))

$$H(X|Y) = \sum_y p(y) H(X|Y = y) \quad (28)$$

$$= - \sum_x \sum_y p(x, y) \log p(x|y) \quad (29)$$

Mutual information (see (9))

$$I(X; Y) = \sum_x \sum_y p(x, y) \log \frac{p(x, y)}{p(x) p(y)} \quad (30)$$

Note that when X and Y are independent, Y can tell us nothing about X and then $I(X; Y) = 0$ in this case.

Maximum entropy models

- **Goal:** estimate p
- Choose p with maximum entropy (or “uncertainty” subject to some constraints)
- **Entropy** is a mathematic measure of the uniformity (uncertainty) of a distribution $p(x, y)$ (see (2))

$$H(p) = - \sum_{x,y} p(x, y) \log p(x, y) \quad (31)$$

- For a conditional distribution $p(y|x)$ its conditional entropy is (see (4))

$$H(p) = - \sum_{x,y} \tilde{p}(x) p(y|x) \log p(y|x) \quad (32)$$

Maximum entropy models

- Collect (x, y) pairs from training data:
 - y : label to be predicted
 - x : the context
- Learn the probability p of each (x, y)
- Maximum entropy

Model all that is known and assume nothing about what is unknown

- to satisfy a set of constraints
- to assume the most “uniform” distribution

Maximum entropy models

Example 1: MT

“We wish to model an expert translator’s decisions concerning the proper French rendering of the English word **in**”

Let p our model of the translator’s decisions and let (x, y) a set of samples: $\{(in, dans), (in, en), \dots\}$

“Suppose that the expert translator always chooses among the following five French phrases: $\{dans, en, \grave{a}, au\ cours\ de, pendant\}$ ”

$$p(dans) + p(en) + p(\grave{a}) + p(au\ cours\ de) + p(pendant) = 1$$

Maximum entropy models

“Suppose we notice that the expert chose either *dans* or *en* 30% of the time”

$$p(\textit{dans}) + p(\textit{en}) = 3/10$$

$$p(\textit{dans}) + p(\textit{en}) + p(\textit{\grave{a}}) + p(\textit{au cours de}) + p(\textit{pendant}) = 1$$

“...in half the cases, the expert chose either *dans* or *\grave{a}*”

$$p(\textit{dans}) + p(\textit{en}) = 3/10$$

$$p(\textit{dans}) + p(\textit{en}) + p(\textit{\grave{a}}) + p(\textit{au cours de}) + p(\textit{pendant}) = 1$$

$$p(\textit{dans}) + p(\textit{\grave{a}}) = 1/2$$

Maximum entropy models

Example 2: POS tagging

- Let say we have the following event space

NN	NNS	NNP	NNPS	VBZ	VBD
----	-----	-----	------	-----	-----

- Empirical data

3	5	11	13	3	1
---	---	----	----	---	---

- Maximize the entropy

$1/e$	$1/e$	$1/e$	$1/e$	$1/e$	$1/e$
-------	-------	-------	-------	-------	-------

- $E[\text{NN}, \text{NNS}, \text{NNP}, \text{NNPS}, \text{VBZ}, \text{VBD}] = 1$

$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$
-------	-------	-------	-------	-------	-------

Maximum entropy models

- N^* are more common than V^* . Add feature $f_N = \{NN, NNS, NNP, NNPS\}$ with $E[f_N] = 32/36$

NN	NNS	NNP	NNPS	VBZ	VBD
8/36	8/36	8/36	8/36	2/36	2/36

- Proper nouns are more frequent than common nouns. Add feature $f_P = \{NNP, NNPS\}$ with $E[f_P] = 24/36$

NN	NNS	NNP	NNPS	VBZ	VBD
4/36	4/36	12/36	12/36	2/36	2/36

Maximum entropy models

Problem: Construct a stochastic model that accurately represents the behaviour of a random process: $p(y|x)$.

Training data

Given a training sample (x, y) , its empirical probability distribution $\tilde{p}$ is defined by:

$$\tilde{p}(x, y) \equiv \frac{1}{N} \times \text{number of times that } (x, y) \text{ occurs in the sample}$$

Features

If *April* is the word following *in*, then the translation of *in* is *en* with frequency 9/10.

$$f(x, y) = \begin{cases} 1 & \text{if } y = \textit{en} \text{ and } \textit{April} \text{ follows } \textit{in} \\ 0 & \text{otherwise} \end{cases} \quad (33)$$

Maximum entropy models

Constraints

Expected value of f with respect to $\tilde{p}(x, y)$:

$$\tilde{p}(f) \equiv \sum_{x,y} \tilde{p}(x, y) f(x, y) \quad (34)$$

Expected value of f with respect to the model $p(y|x)$:

$$p(f) \equiv \sum_{x,y} \tilde{p}(x) p(y|x) f(x, y) \quad (35)$$

$$\sum_{x,y} \tilde{p}(x) p(y|x) f(x, y) = \sum_{x,y} \tilde{p}(x, y) f(x, y) \quad (36)$$

Maximum entropy models

Given n features f_i , we would like p lie in the subset $\mathcal{C}$ of $\mathcal{P}$ defined by

$$\mathcal{C} \equiv \{p \in \mathcal{P} \mid p(f_i) = \tilde{p}(f_i) \text{ for } i \in \{1, 2, \dots, n\}\} \quad (37)$$

Conditional entropy

$$H(p) = - \sum_{x,y} \tilde{p}(x) p(y|x) \log p(y|x) \quad (38)$$

To select a model from a set $\mathcal{C}$ of allowed probability distributions, choose the model $p_* \in \mathcal{C}$ with maximum entropy $H(p)$:

$$p_* = \arg \max_{p \in \mathcal{C}} H(p) \quad (39)$$

Maximum entropy models

Solution to the primal problem:

<http://www.cs.cmu.edu/afs/cs/user/abberger/www/html/tutorial/node7.html#SECTION00024000000000000000>

The Maximum entropy solution p_* have the form:

$$p(y|x) = \frac{1}{Z(x)} \exp \left(\sum_{i=1}^k \lambda_i f_i(x, y) \right) \quad (40)$$

$$Z(x) = \sum_y \exp \left(\sum_{i=1}^k \lambda_i f_i(x, y) \right) \quad (41)$$

where k is the number of features

Let λ^* be

$$\lambda^* = \arg \max_{\lambda} \frac{1}{Z(x)} \exp \left(\sum_{i=1}^k \lambda_i f_i(x, y) \right) \quad (42)$$

Then $p_{\lambda^*} = p_*$

Maximum entropy models

Conditional likelihood of the training data (X, Y)

$$\log p(Y|X, \lambda) = \log \prod_{(x,y) \in (X,Y)} p(y|x, \lambda) \quad (43)$$

$$= \sum_{(x,y) \in (X,Y)} \log \frac{\exp \left(\sum_{i=1}^k \lambda_i f_i(x, y) \right)}{\sum_y \exp \left(\sum_{i=1}^k \lambda_i f_i(x, y) \right)} \quad (44)$$

$$= \sum_{(x,y) \in (X,Y)} \sum_{i=1}^k \lambda_i f_i(x, y) - \sum_{(x,y) \in (X,Y)} \log \sum_y \exp \left(\sum_{i=1}^k \lambda_i f_i(x, y) \right) \quad (45)$$

Solution: Compute each derivative independently and equal to 0

Maximum entropy models

$$\frac{\partial}{\partial \lambda_i} \sum_{(x,y) \in (X,Y)} \sum_{i=1}^k \lambda_i f_i(x,y)} = \sum_{(x,y) \in (X,Y)} f_i(x,y) \quad (46)$$

$$\frac{\partial}{\partial \lambda_i} \sum_{(x,y) \in (X,Y)} \log \sum_y \exp \left(\sum_{i=1}^k \lambda_i f_i(x,y) \right) \quad (47)$$

$$= \sum_{(x,y) \in (X,Y)} \frac{1}{\sum_{y''} \exp \left(\sum_{i=1}^k \lambda_i f_i(x,y'') \right)} \sum_{y'} \exp \left(\sum_{i=1}^k \lambda_i f_i(x,y') \right) \frac{\partial \sum_{i=1}^k \lambda_i f_i(x,y')}{\partial \lambda_i} \quad (48)$$

$$= \sum_{(x,y) \in (X,Y)} \sum_{y'} \frac{\exp \left(\sum_{i=1}^k \lambda_i f_i(x,y') \right)}{\sum_{y''} \exp \left(\sum_{i=1}^k \lambda_i f_i(x,y'') \right)} \frac{\partial \sum_{i=1}^k \lambda_i f_i(x,y')}{\partial \lambda_i} = \sum_{(x,y) \in (X,Y)} \sum_{y'} p_{\lambda}(y' | x) f_i(x,y') \quad (49)$$

The optimum is achieved when (see Eq. (36)):

$$\sum_{(x,y) \in (X,Y)} f_i(x,y) = \sum_{(x,y) \in (X,Y)} \sum_{y'} p_{\lambda}(y' | x) f_i(x,y') \quad (50)$$

Maximum entropy models

Solution to the dual problem: IIS algorithm Details [here](#) and [here](#)

1. Start with some (arbitrary) value for each λ_i
2. Repeat until convergence:

$$(a) \text{ Solve } \frac{\partial \mathcal{B}(\delta)}{\partial \delta_i} = 0 \quad \text{for } \delta_i \quad (51)$$

$$(b) \text{ Set } \lambda_i = \lambda_i + \delta_i$$

where:

$$\frac{\partial \mathcal{B}(\delta)}{\partial \delta_i} = \overbrace{\sum_{x,y} \tilde{p}(x,y) f_i(x,y)}^{\tilde{p}(f_i)} - \overbrace{\sum_{x,y} \tilde{p}(x) p_\lambda(y|x) f_i(x,y)}^{p_\lambda(f_i)} e^{\delta_i f^\#(x,y)} \quad (52)$$

If $f^\#(x,y) = M$ then

$$\delta_i = \frac{1}{M} \log \frac{\tilde{p}(f_i)}{p_\lambda(f_i)} \quad (53)$$

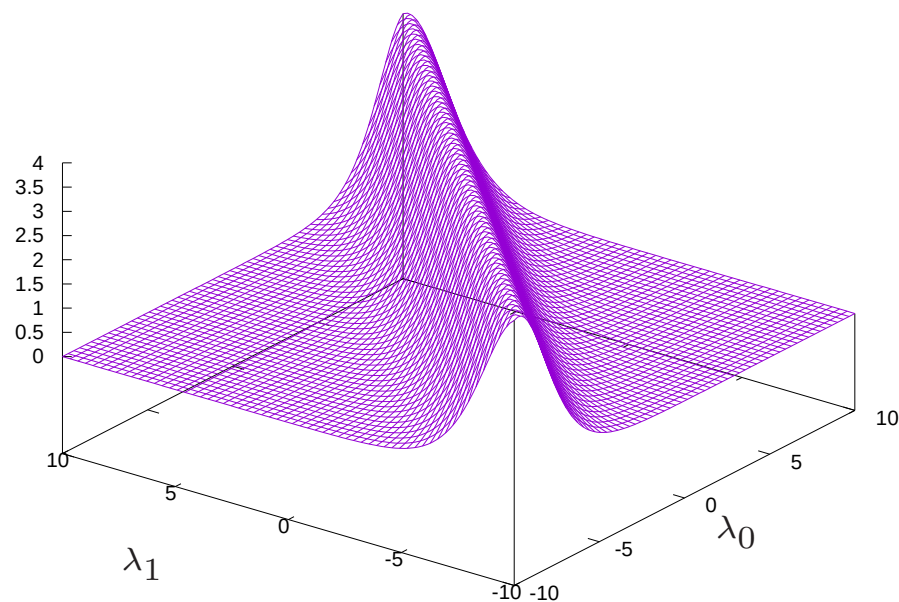
Maximum entropy models

Example to illustrate the optimization problem (just two features)

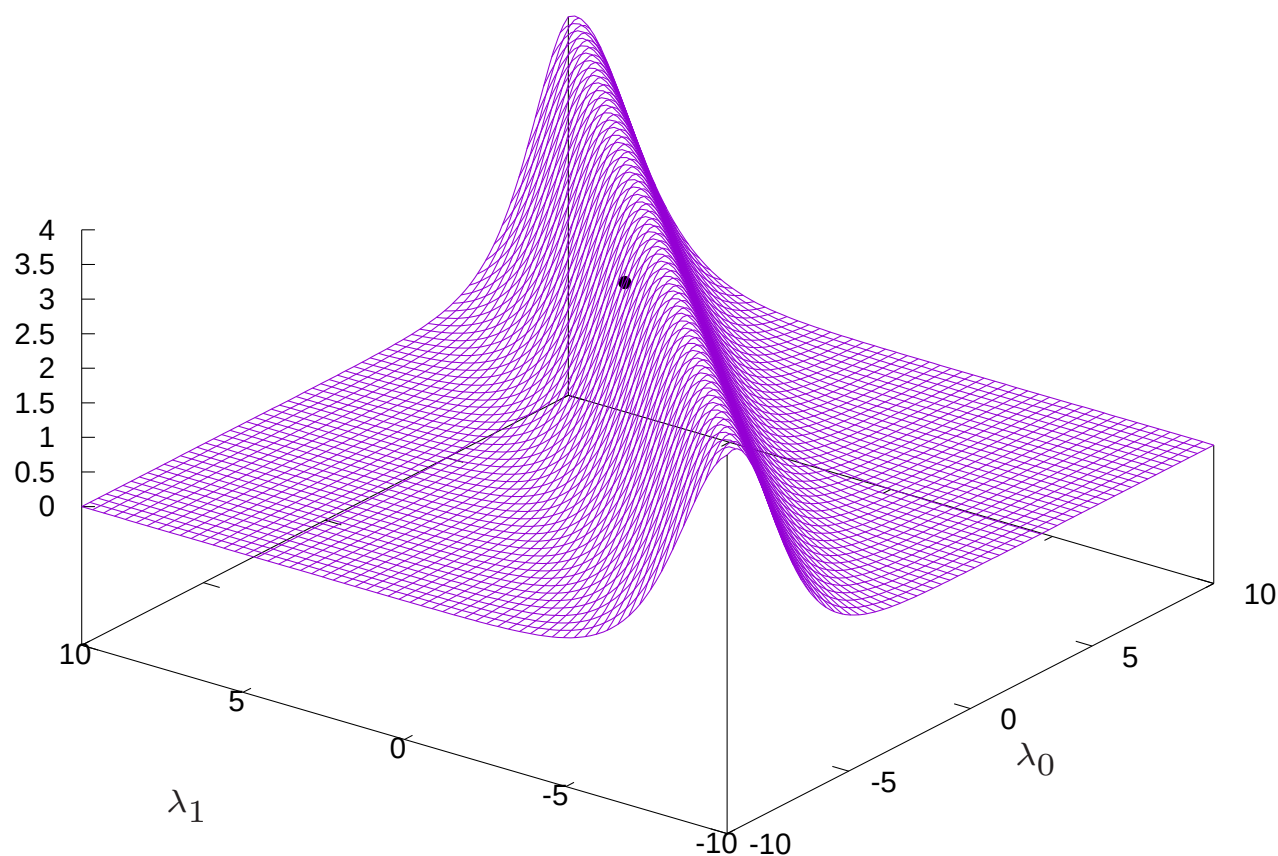
$$C = \{(w_0, c_0), (w_0, c_0), (w_0, c_0), (w_1, c_1), (w_1, c_1), (w_1, c_1), (w_1, c_1), (w_1, c_1), (w_1, c_1), (w_1, c_1)\}$$

$$\lambda_0 = f(w_0, c_0) \quad \lambda_1 = f(w_1, c_1)$$

$H(p)$ with $\lambda_0 = 0, \lambda_1 = 0$



$H(p)$ with $\lambda_0 = 0, \lambda_1 = 0 \rightarrow H(p) = 2.84, \lambda_0 = .69, \lambda_1 = .69$



Maximum entropy models. Smoothing

- Lots of features
- Lots of sparsity
- Overfitting very easy
- Many features at test time could not be seen in training time
- Feature weights can be infinite and optimizer can take a long time to get to those infinities
- Smoothing is needed!

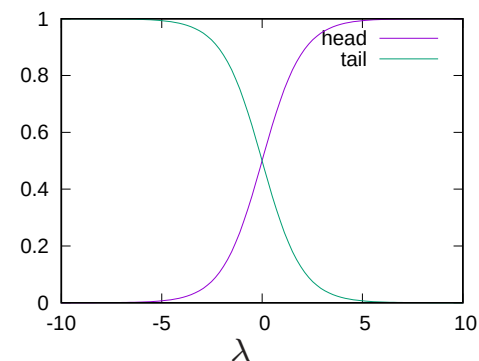
Maximum entropy models. Smoothing

- Assume the following empirical distribution: a coin with 2 faces and identical probability
- Features: head, tail
- Model distribution

$$p_h = \frac{e^{\lambda_h}}{e^{\lambda_h} + e^{\lambda_t}} \quad p_t = \frac{e^{\lambda_t}}{e^{\lambda_h} + e^{\lambda_t}}$$

- Only one degree of freedom: $\lambda = \lambda_h - \lambda_t$

$$p_h = \frac{e^{\lambda_h} e^{\lambda-t}}{e^{\lambda_h} e^{\lambda-t} + e^{\lambda_t} e^{\lambda-t}} = \frac{e^{\lambda}}{e^{\lambda} + e^0} = \frac{e^{\lambda}}{e^{\lambda} + 1} \quad p_t = \frac{1}{1 + e^{\lambda}}$$

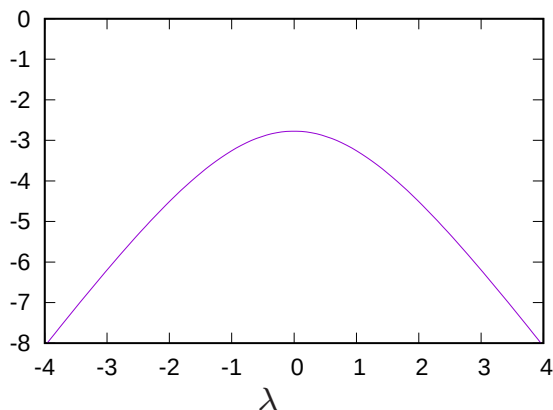


Maximum entropy models. Smoothing

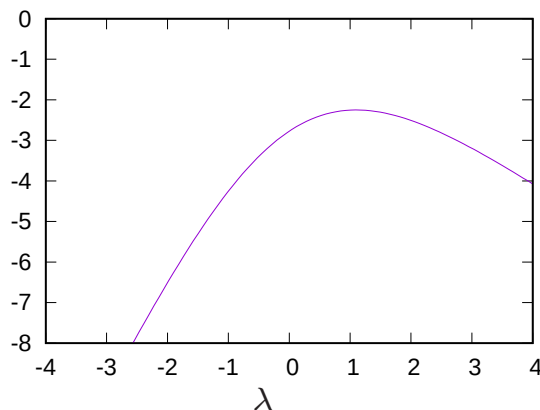
- For a sample with h and t , the log likelihood is:

$$\begin{aligned}h \log p_h + t \log p_t &= h(\lambda - \log(1 + e^\lambda)) + t(\log 1 - \log(1 + e^\lambda)) \\&= h\lambda - h \log(1 + e^\lambda) + t \log(1 + e^\lambda) \\&= h\lambda - (t + h) \log(1 + e^\lambda)\end{aligned}$$

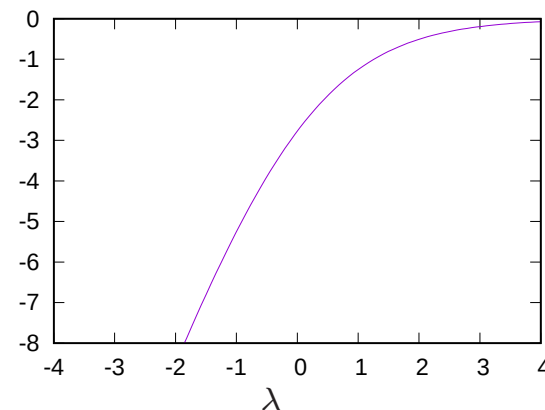
$h = 2 \quad t = 2$



$h = 3 \quad t = 1$



$h = 4 \quad t = 0$



- λ goes to $\infty \longrightarrow$ long time to convergence

Maximum entropy models. Smoothing

- To include a prior on the parameter values and change the optimization objective:

$$\log p(Y, \lambda \mid X) \approx \log p(\lambda) + \log p(Y \mid X, \lambda)$$

- Gaussian prior: each parameter will be distributed according to a gaussian mean μ and variance σ^2 .
- Usually $\mu = 0$ and $2\sigma^2 = 1$
- New optimization function

$$\log p(Y, \lambda \mid X) = \sum_{(x,y) \in (X,Y)} p(y \mid x, \lambda) - \sum_i \frac{(\lambda_i - \mu_i)^2}{2\sigma_i^2} + k \quad (54)$$

- Change the derivative

$$\sum_{(x,y) \in (X,Y)} f_i(x, y) - \sum_{(x,y) \in (X,Y)} \sum_{y'} p_\lambda(y' \mid x) f_i(x, y') - \frac{(\lambda_i - \mu_i)}{\sigma_i^2} \quad (55)$$

Maximum entropy models

Exercise 1. Given a set of words, such that each word has associated a class label c_0 or c_1 :

$$C = \{(w_0, c_0), (w_0, c_0), (w_1, c_0), (w_1, c_0), (w_1, c_0), (w_1, c_0), \\ (w_0, c_1), (w_2, c_1), (w_2, c_1), (w_2, c_1)\}$$

Obtain a maximum entropy model for classifying a text containing those words. The following features have been defined for this purpose:

$$f(x, y) = \begin{cases} 1 & \text{if } y = c_i \text{ and it has the word } y = w_j \\ 0 & \text{otherwise} \end{cases}$$

1. Suppose that all λ_i are initialized to 0, then compute the increasing δ_0 for the feature $f_0 = f(w_0, c_0)$ with IIS algorithm.
2. Suppose that a ME model has been learned and the following values have been obtained:

$$\begin{aligned} \lambda_0 &= 0.3 \text{ associated to the feature } f_0 = f(w_0, c_0), \\ \lambda_1 &= 3.0 \text{ associated to the feature } f_1 = f(w_1, c_0), \\ \lambda_2 &= -0.4 \text{ associated to the feature } f_2 = f(w_0, c_1), \\ \lambda_3 &= 3.0 \text{ associated to the feature } f_3 = f(w_2, c_1). \end{aligned}$$

Compute in which class is classified a text if it contains the word w_0 .

Maximum entropy models

Solution

a)

$$\tilde{p}(w_0, c_0) = \frac{2}{10} = \frac{1}{5}, \quad \tilde{p}(w_0) = \frac{3}{10} \quad \text{y} \quad p_\lambda(c_0|w_0) = \frac{1}{2}$$

Then:

$$\delta_0 = \log \frac{\frac{1}{5}}{\frac{3}{10} \frac{1}{2}} = \log \frac{4}{3} = 0.288$$

b)

$$p_\lambda(c_0|w_0) = \frac{\exp(0.3)}{\exp(0.3) + \exp(-0.4)} = 0.668$$

$$p_\lambda(c_1|w_0) = \frac{\exp(-0.4)}{\exp(0.3) + \exp(-0.4)} = 0.332$$

and therefore that sample is classified in class c_0 .

Exercise (*): Suppose that all λ_i are initialized to 0, then compute the increasing δ_{21} for the feature $f_{21} = f(w_2, c_1)$ with the IIS algorithm.

Compute in which class is classified a text if it contains the words w_0, w_0, w_2 .

Maximum entropy models

Exercise 2. Let a classification problem into three classes A , B and C , where the classification is carried out with a Maximum Entropy classifier. The classification of each samples is performed according to 3 features out of 5, noted as $(c_0, c_1, c_2, c_3, c_4)$.

The features of the Maximum Entropy classifier are defined as:

$$f(y, x) = \begin{cases} 1 & \text{if } y = S \text{ and the feature } c_j \text{ is active in } x \\ 0 & \text{otherwise} \end{cases}$$

where $S \in \{A, B, C\}$.

Suppose that a model has been trained and the obtained parameters are:

$$\begin{aligned} \lambda_{A,c_0} &= 0.37 & \lambda_{A,c_1} &= 0.0 & \lambda_{A,c_2} &= -0.04 & \lambda_{A,c_3} &= 0.08 & \lambda_{A,c_4} &= 0.0 \\ \lambda_{B,c_0} &= 0.0 & \lambda_{B,c_1} &= 0.0 & \lambda_{B,c_2} &= -0.04 & \lambda_{B,c_3} &= -0.05 & \lambda_{B,c_4} &= -0.28 \\ \lambda_{C,c_0} &= 0.0 & \lambda_{C,c_1} &= 0.23 & \lambda_{C,c_2} &= 0.06 & \lambda_{C,c_3} &= -0.05 & \lambda_{C,c_4} &= 0.32 \end{aligned}$$

Compute the class of a sample if the features c_0 , c_2 and c_3 has been observed.

Maximum entropy models

Solution

$$p_{\lambda}(A|(c_0, c_2, c_3)) = \frac{\exp(0.37 - 0.04 + 0.08)}{\exp(0.37 - 0.04 + 0.0) + \exp(0.0 - 0.04 - 0.05) + \exp(0.0 + 0.06 - 0.0)} = 0.43$$

$$p_{\lambda}(B|(c_0, c_2, c_3)) = \frac{\exp(0.0 - 0.04 - 0.05)}{\exp(0.37 - 0.04 + 0.0) + \exp(0.0 - 0.04 - 0.05) + \exp(0.0 + 0.06 - 0.0)} = 0.27$$

$$p_{\lambda}(C|(c_0, c_2, c_3)) = \frac{\exp(0.0 + 0.06 - 0.05)}{\exp(0.37 - 0.04 + 0.0) + \exp(0.0 - 0.04 - 0.05) + \exp(0.0 + 0.06 - 0.0)} = 0.30$$

and therefore the samples is classified in class A .

Maximum entropy models

Available ME tools:

- MALLET: <http://mallet.cs.umass.edu/>
- NLTK: <http://www.nltk.org/>
- WEKA: <http://www.cs.waikato.ac.nz/ml/weka/>

Recommended web pages:

- <http://www.cs.cmu.edu/afs/cs/user/abberger/www/html/tutorial/tutorial.html>
- http://en.wikipedia.org/wiki/Principle_of_maximum_entropy
- <http://www.inference.phy.cam.ac.uk/hmw26/crf/>

Exercise ():** Choose a toolkit and a dataset and develop a classifier trained with that toolkit. The mark on this exercise will depend on the obtained results compared with other state-of-the-art results.

Exercise ():** Consider the set of sentences: $S = \{("a a", C_0), ("b b", C_0), ("a b", C_1), ("b a", C_1)\}$. Compute a Maximum Entropy classifier with features $f_{C_i, x}$ with $i \in \{1, 2\}, x \in \{a, b\}$ and compute the classification error. Describe carefully the conclusion that you get and justify them.

Exercise ():** Modify mallet to compute (55) and report experiments on a dataset.

Maximum entropy models

Feature selection: Information Gain. The idea is to use the Mutual Information for choosing the features

$$I(X;Y) = H(Y) - H(Y|X) = H(Y) - \sum_x p(X=x)H(Y|X=x)$$

Example. Suppose that we have two type of texts A and B , and each text has only one word. We have 6 instances of text of type A : $\{a, a, a, a, b, c\}$; and 4 instances of text of type B : $\{a, b, b, c\}$.

$$H(Y) = - \sum_{A,B} p(Y) \log p(Y) = -3/5 \log 3/5 - 2/5 \log 2/5 = .97 \text{ bits}$$

$$p(X=a) = \frac{1}{2} \quad p(A|a) = \frac{p(A,a)}{p(a)} = \frac{4}{5} \quad p(B|a) = \frac{p(B,a)}{p(a)} = \frac{1}{5}$$

$$p(X=b) = \frac{3}{10} \quad p(A|b) = \frac{p(A,b)}{p(b)} = \frac{1}{3} \quad p(B|b) = \frac{p(B,b)}{p(b)} = \frac{2}{3}$$

$$p(X=c) = \frac{1}{5} \quad p(A|c) = \frac{p(A,c)}{p(c)} = \frac{1}{2} \quad p(B|c) = \frac{p(B,c)}{p(c)} = \frac{1}{2}$$

Maximum entropy models

The word can be classified according to entropy $H(Y)$ reduction:

$$\begin{aligned} & -p(X = x)(p(Y = A|X = x) \log p(Y = A|X = x) \\ & \quad + p(Y = B|X = x) \log p(Y = B|X = x)) \end{aligned}$$

$$X = a \quad \longrightarrow \quad -\frac{1}{2} \left(\frac{4}{5} \log \frac{4}{5} + \frac{1}{5} \log \frac{1}{5} \right) = .25$$

$$X = b \quad \longrightarrow \quad -\frac{3}{10} \left(\frac{1}{3} \log \frac{1}{3} + \frac{2}{3} \log \frac{2}{3} \right) = .19$$

$$X = c \quad \longrightarrow \quad -\frac{1}{5} \left(\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2} \right) = .13$$

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2. Entropy measures for grammatical models
3. Maximum entropy models
4. **Regularized EM algorithm**
5. Discriminative training criterion
6. Semi-supervised learning
7. Active learning

EM algorithm

Goal: find θ such that $\mathcal{P}(\mathbf{X}|\theta)$ is maximum (Maximum Likelihood estimate for θ)

Log likelihood function:

$$L(\theta) = \ln \mathcal{P}(\mathbf{X}|\theta). \quad (56)$$

Suppose that after iteration n^{th} the current parameters are θ_n and we want to estimate these parameters. Then, we want to compute an updated estimate θ such that,

$$L(\theta) > L(\theta_n). \quad (57)$$

or to maximize the difference

$$L(\theta) - L(\theta_n) = \ln \mathcal{P}(\mathbf{X}|\theta) - \ln \mathcal{P}(\mathbf{X}|\theta_n). \quad (58)$$

EM algorithm

Denote the hidden random vector by $\mathbf{Z}$ and a given realization by $\mathbf{z}$:

$$\mathcal{P}(\mathbf{X}|\theta) = \sum_{\mathbf{z}} \mathcal{P}(\mathbf{X}, \mathbf{z}|\theta) = \sum_{\mathbf{z}} \mathcal{P}(\mathbf{X}|\mathbf{z}, \theta) \mathcal{P}(\mathbf{z}|\theta). \quad (59)$$

Then, Equation (58) can be rewritten as:

$$L(\theta) - L(\theta_n) = \ln \left(\sum_{\mathbf{z}} \mathcal{P}(\mathbf{X}|\mathbf{z}, \theta) \mathcal{P}(\mathbf{z}|\theta) \right) - \ln \mathcal{P}(\mathbf{X}|\theta_n). \quad (60)$$

EM algorithm

Then, from Equation (7):

$$L(\theta) - L(\theta_n) = \ln \left(\sum_{\mathbf{z}} \mathcal{P}(\mathbf{X}|\mathbf{z}, \theta) \mathcal{P}(\mathbf{z}|\theta) \right) - \ln \mathcal{P}(\mathbf{X}|\theta_n) \quad (61)$$

$$= \ln \left(\sum_{\mathbf{z}} \mathcal{P}(\mathbf{X}|\mathbf{z}, \theta) \mathcal{P}(\mathbf{z}|\theta) \frac{\mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n)}{\mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n)} \right) - \ln \mathcal{P}(\mathbf{X}|\theta_n) \quad (62)$$

$$= \ln \left(\sum_{\mathbf{z}} \mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n) \frac{\mathcal{P}(\mathbf{X}|\mathbf{z}, \theta) \mathcal{P}(\mathbf{z}|\theta)}{\mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n)} \right) - \ln \mathcal{P}(\mathbf{X}|\theta_n) \quad (63)$$

$$\geq \sum_{\mathbf{z}} \mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n) \ln \left(\frac{\mathcal{P}(\mathbf{X}|\mathbf{z}, \theta) \mathcal{P}(\mathbf{z}|\theta)}{\mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n)} \right) - \ln \mathcal{P}(\mathbf{X}|\theta_n) \quad (64)$$

$$= \sum_{\mathbf{z}} \mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n) \ln \left(\frac{\mathcal{P}(\mathbf{X}|\mathbf{z}, \theta) \mathcal{P}(\mathbf{z}|\theta)}{\mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n)} \right) - \sum_{\mathbf{z}} \mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n) \ln \mathcal{P}(\mathbf{X}|\theta_n) \quad (65)$$

$$= \sum_{\mathbf{z}} \mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n) \ln \left(\frac{\mathcal{P}(\mathbf{X}|\mathbf{z}, \theta) \mathcal{P}(\mathbf{z}|\theta)}{\mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n) \mathcal{P}(\mathbf{X}|\theta_n)} \right) \quad (66)$$

$$\triangleq \Delta(\theta|\theta_n) \quad (67)$$

EM algorithm

Then:

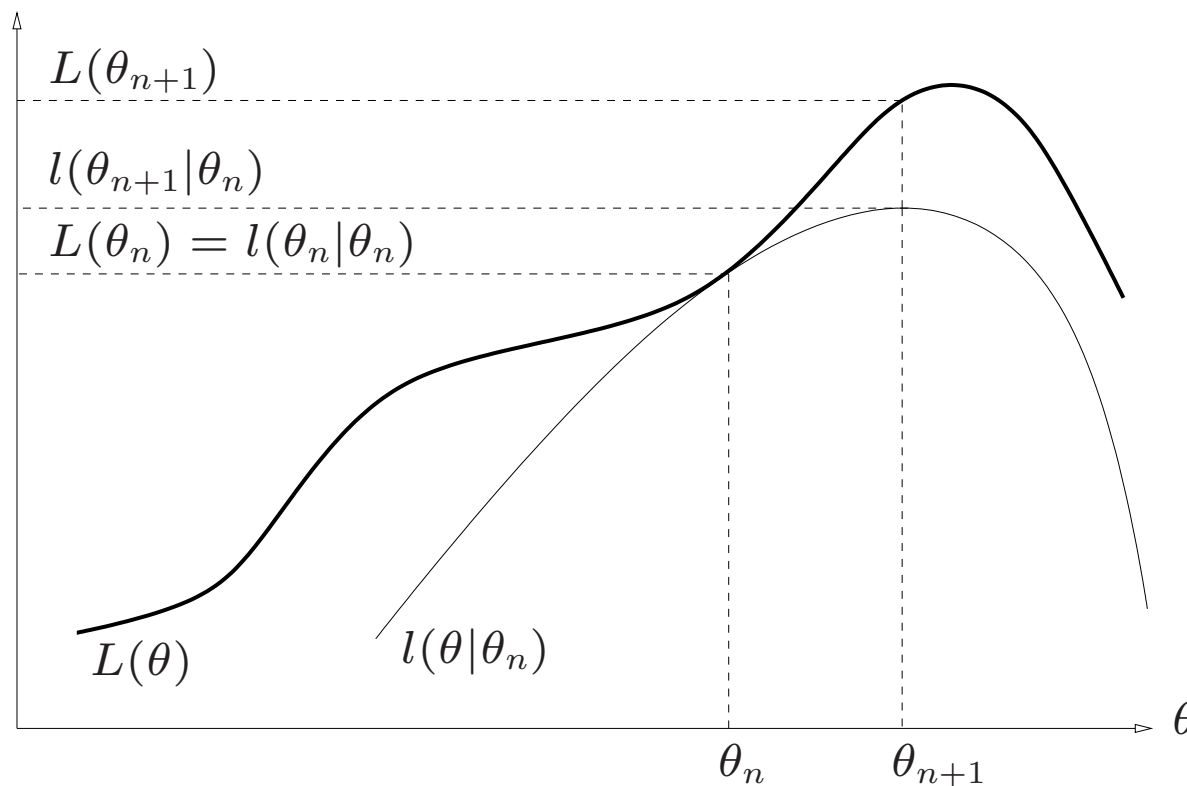
$$L(\theta) \geq L(\theta_n) + \Delta(\theta|\theta_n).$$

If we define:

$$l(\theta|\theta_n) \triangleq L(\theta_n) + \Delta(\theta|\theta_n), \quad \text{note that } L(\theta_n) \text{ is a known value}$$

then

$$L(\theta) \geq l(\theta|\theta_n).$$



EM algorithm

Since $l(\theta|\theta_n)$ is bounded above by $L(\theta)$, any θ that increases $l(\theta|\theta_n)$ will also increase $L(\theta)$.

$$\theta_{n+1} = \arg \max_{\theta} \{l(\theta|\theta_n)\} \quad (68)$$

$$= \arg \max_{\theta} \left\{ L(\theta_n) + \sum_{\mathbf{z}} \mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n) \ln \frac{\mathcal{P}(\mathbf{X}|\mathbf{z}, \theta) \mathcal{P}(\mathbf{z}|\theta)}{\mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n) \mathcal{P}(\mathbf{X}|\theta_n)} \right\} \quad (69)$$

$$= \arg \max_{\theta} \left\{ \sum_{\mathbf{z}} \mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n) \ln \mathcal{P}(\mathbf{X}|\mathbf{z}, \theta) \mathcal{P}(\mathbf{z}|\theta) \right\} \quad (70)$$

$$= \arg \max_{\theta} \left\{ \sum_{\mathbf{z}} \mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n) \ln \mathcal{P}(\mathbf{X}, \mathbf{z}|\theta) \right\} \quad (71)$$

$$= \arg \max_{\theta} \left\{ \mathbf{E}_{\mathbf{Z}|\mathbf{X}, \theta_n} \{ \ln \mathcal{P}(\mathbf{X}, \mathbf{z}|\theta) \} \right\} \quad (72)$$

1. *E-step*: Determine $\mathcal{P}(\mathbf{Z}|\mathbf{X}, \theta_n)$
2. *M-step*: Maximize (72) with respect to θ .

EM algorithm: example with a Gaussian mixture

Simplification: unidimensional Gaussian mixture

$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}) \quad (73)$$

Let suppose that $\mathbf{z}$ is a K -dimensional binary random variable where some value z_k is equal to 1 and all the other values are equal to 0. Variable $\mathbf{z}$ has K possible states.

$$p(\mathbf{z}_k = 1) = \pi_k, \quad \sum_{k=1}^K \pi_k = 1$$

Alternative notation:

$$p(\mathbf{z}) = \prod_{k=1}^K \pi_k^{z_k}$$

In addition

$$p(\mathbf{x} | \mathbf{z}_k = 1) = p(\mathbf{x} | \mu_k, \sigma_k)$$

or

$$p(\mathbf{x} | \mathbf{z}) = \prod_{k=1}^K p(\mathbf{x} | \mu_k, \sigma_k)^{z_k}$$

EM algorithm: example with a Gaussian mixture

In this way:

$$\begin{aligned} p(\mathbf{x}) &= \sum_{\mathbf{z}} p(\mathbf{z})p(\mathbf{x}|\mathbf{z}) = \sum_{k=1}^K \pi_k p(\mathbf{x}|\mu_k, \sigma_k) \\ &= \sum_{k=1}^K \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} e^{-\frac{(\mathbf{x}-\mu_k)^2}{2\sigma_k^2}} \end{aligned} \quad (74)$$

Given a sample S , the log likelihood is defined as:

$$L_S(\theta) = \ln \prod_{m=1}^M p(\mathbf{x}_m|\theta) \quad (75)$$

$$= \sum_{m=1}^M \ln \sum_{k=1}^K \pi_k p(\mathbf{x}_m|\mu_k, \sigma_k) \quad (76)$$

EM algorithm: example with a Gaussian mixture

Simplification: unidimensional Gaussian mixture with unknown parameters π_k . The other parameters are all known and constant along all iterations (see [Bishop 2006] pp. 430–443).

1. *E-step*:

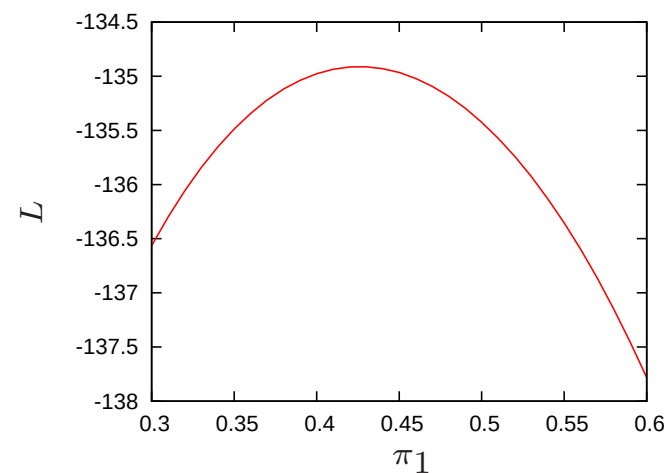
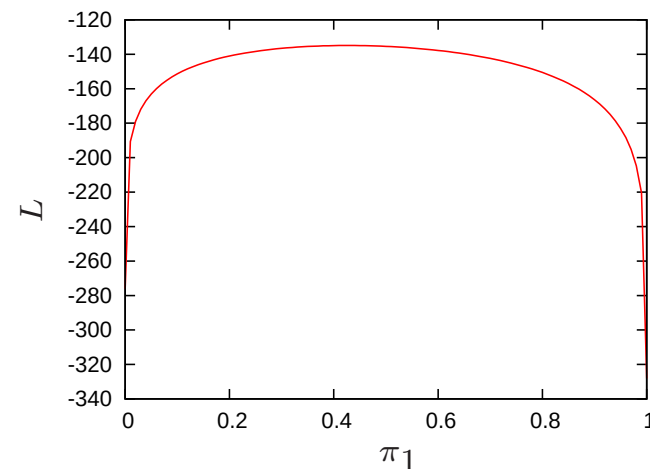
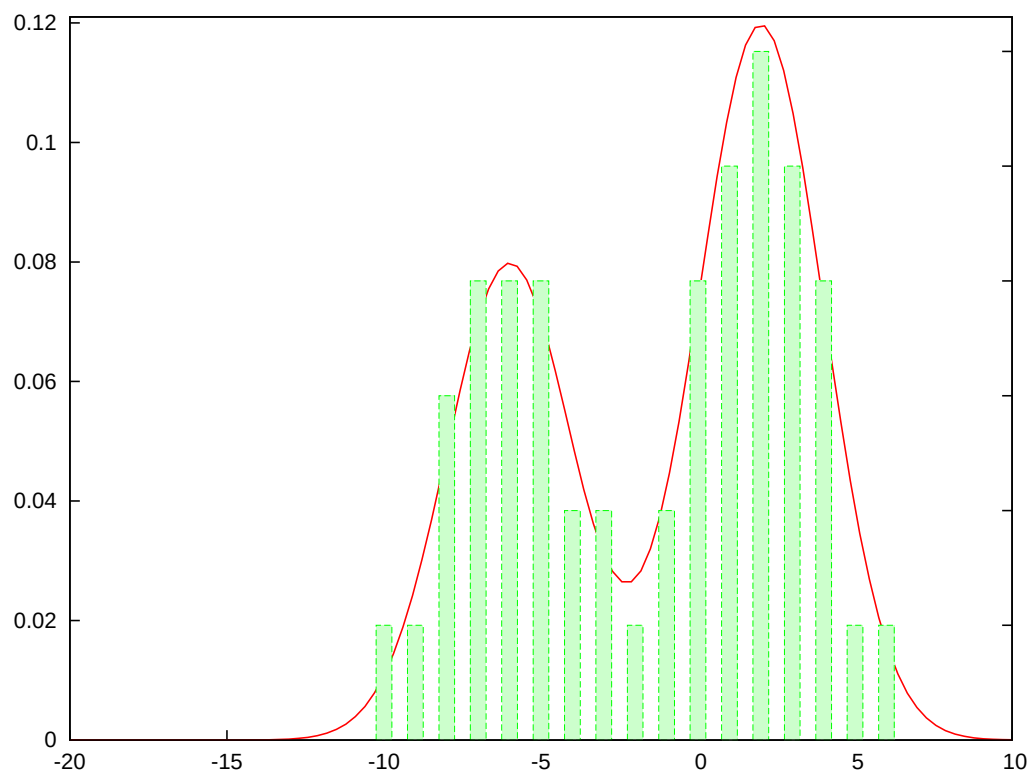
$$\hat{z}_{km} = \frac{\pi_k^{(t)} p(\mathbf{x}_m | \theta_k)}{\sum_{k'} \pi_{k'}^{(t)} p(\mathbf{x}_m | \theta_{k'})} \quad (77)$$

2. *M-step*:

$$\pi_k^{(t+1)} = \frac{\sum_{m=1}^M \hat{z}_{km}}{M} \quad (78)$$

EM algorithm: example with a Gaussian mixture

Let a gaussian mixture of two unidimensional distributions, with known mean (-6 and 2) and equal and known variance (4) where only π_1, π_2 (0.4 and 0.6) are unknown. Let suppose that a training sample of 50 unidimensional vectors is available:



EM algorithm: example with a Gaussian mixture

$$\theta^{(0)} = (\pi_1 = 0.700, \pi_2 = 0.300)$$

$$\theta^{(1)} = (\pi_1 = 0.445, \pi_2 = 0.555)$$

$$\theta^{(2)} = (\pi_1 = 0.427, \pi_2 = 0.573)$$

$$\theta^{(3)} = (\pi_1 = 0.426, \pi_2 = 0.574)$$

$$\theta^{(4)} = (\pi_1 = 0.426, \pi_2 = 0.574)$$

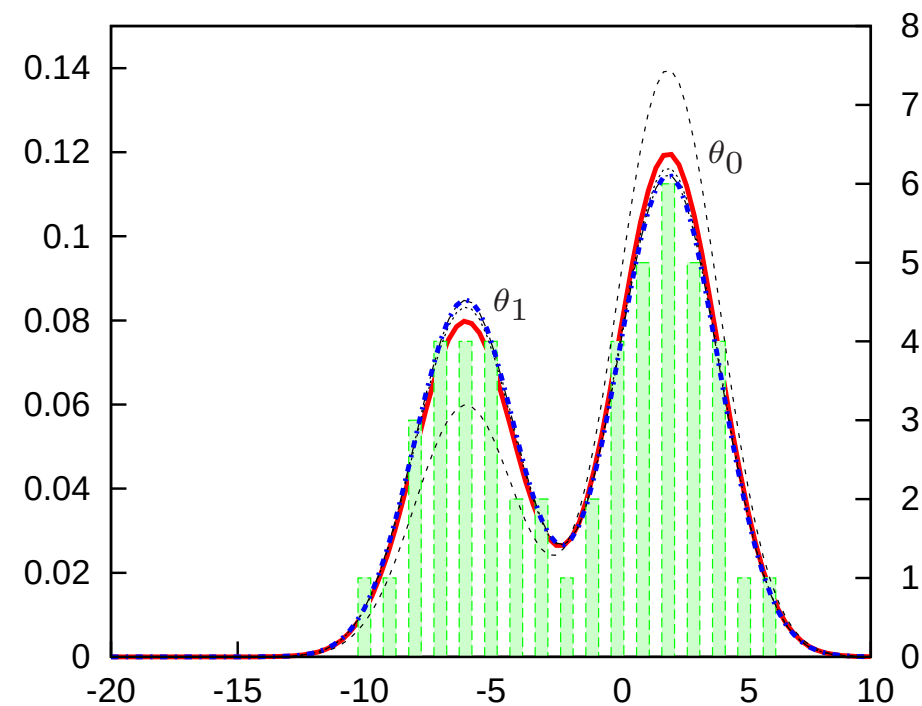
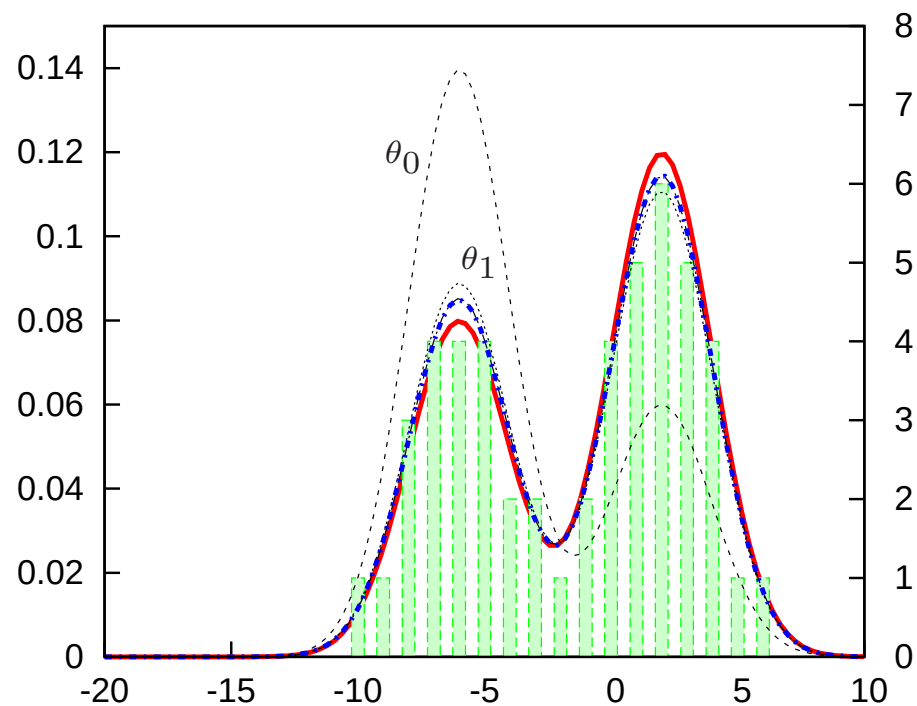
$$\theta^{(0)} = (\pi_1 = 0.300, \pi_2 = 0.700)$$

$$\theta^{(1)} = (\pi_1 = 0.417, \pi_2 = 0.583)$$

$$\theta^{(2)} = (\pi_1 = 0.425, \pi_2 = 0.575)$$

$$\theta^{(3)} = (\pi_1 = 0.426, \pi_2 = 0.574)$$

$$\theta^{(4)} = (\pi_1 = 0.426, \pi_2 = 0.574)$$



EM algorithm: example with a Gaussian mixture

Exercise ():** Reproduce an example similar to the previous example with three unidimensional distributions, with known mean and equal and known variance where only π_1, π_2, π_3 are unknown. But the data have to be generated with two gaussians. Obtain the corresponding plots.

Exercise ():** Reproduce an example similar to the previous example with three unidimensional distributions, with known mean and equal and known variance where only π_1, π_2, π_3 are unknown. The data have to be generated with three gaussians. Obtain the corresponding plots.

Exercise ():** Reproduce an example similar to the previous example with two bidimensional distributions, with known mean vector and equal and known variance vector where only π_1, π_2, π_3 are unknown. Obtain the corresponding plots.

Exercise (*):** Reproduce an example similar to the previous example with two unidimensional distributions, with equal and known variance where the mean of each distribution and π_1, π_2 are unknown. Obtain the corresponding plots.

Exercise (*):** Reproduce an example similar to the previous example with two unidimensional distributions, with equal and known mean where the variances and π_1, π_2 are unknown. Obtain the corresponding plots.

EM algorithm: Regularization

Motivation: To choose a suitable missing data.

“This question is not addressed in the EM algorithm because the likelihood function does not reflect any influence of the missing data.” [Li, 2005]

Approach: *“To regularize the likelihood function with a suitable functional of the distribution of the complete data.”*

Regularized EM:

$$\tilde{L}(\mathbf{X}|\theta) = L(\mathbf{X}|\theta) + \gamma R(\mathbf{X}, \mathbf{Z}|\theta) \quad (79)$$

Ideal situation: the missing data has little uncertainty given the observations
→ Mutual information

Regularization function: Mutual information

$$\tilde{L}(\mathbf{X}|\theta) = L(\mathbf{X}|\theta) + \gamma I(\mathbf{X}, \mathbf{Z}|\theta) \quad (80)$$

If we assume that $\mathbf{Z}$ follows a uniform distribution ($I(\mathbf{X}; \mathbf{Z}) = H(\mathbf{Z}) - H(\mathbf{Z}|\mathbf{X})$):

$$\tilde{L}(\mathbf{X}|\theta) = L(\mathbf{X}|\theta) - \gamma H(\mathbf{Z}|\mathbf{X}; \theta) \quad (81)$$

EM algorithm: Regularization

For a Gaussian mixture:

$$\begin{aligned} \tilde{L}(\theta; \mathbf{X}) &= \sum_{m=1}^M \ln \sum_{k=1}^K \pi_k p(\mathbf{x}_m | \mu_k, \sigma_k) \\ &+ \underbrace{\gamma \int \sum_{k=1}^K \frac{\pi_k p(\mathbf{x} | \mu_k, \sigma_k)}{p(\mathbf{x} | \mu, \sigma)} \ln \left(\frac{\pi_k p(\mathbf{x} | \mu_k, \sigma_k)}{p(\mathbf{x} | \mu, \sigma)} \right) p(\mathbf{x} | \mu, \sigma) d\mathbf{x}}_{\text{See (4)}} \end{aligned} \quad (82)$$

$$\begin{aligned} \tilde{Q}(\theta | \theta^{(t)}) &= \sum_{k=1}^K \sum_{m=1}^M p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}) \ln \pi_k \\ &+ \sum_{k=1}^K \sum_{m=1}^M p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}) \ln p(\mathbf{x}_m | \mu_k, \sigma_k) \quad (\text{See (60)}) \\ &+ \gamma \int \sum_{k=1}^K \frac{\pi_k p(\mathbf{x} | \mu_k, \sigma_k)}{p(\mathbf{x} | \mu, \sigma)} \ln \left(\frac{\pi_k p(\mathbf{x} | \mu_k, \sigma_k)}{p(\mathbf{x} | \mu, \sigma)} \right) p(\mathbf{x} | \mu, \sigma) dx \end{aligned} \quad (83)$$

EM algorithm: Regularization

Introducing a Lagrangian and solving [Li, 2005] (Compare with (77) and (78)):

$$\pi_k^{(t+1)} = \frac{\sum_{m=1}^M p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}) (1 + \gamma \ln p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}))}{\sum_{m=1}^M \sum_{k=1}^K p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}) (1 + \gamma \ln p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}))} \quad (84)$$

where

$$p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}) = \frac{\pi_k^{(t)} p(\mathbf{x}_m | \theta_k^{(t)})}{\sum_{k'} \pi_{k'}^{(t)} p(\mathbf{x}_m | \theta_{k'}^{(t)})} \quad (85)$$

The other parameters:

$$\mu_k^{(t+1)} = \frac{\sum_{m=1}^M \mathbf{x}_m p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}) (1 + \gamma \ln p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}))}{\sum_{m=1}^M p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}) (1 + \gamma \ln p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}))} \quad (86)$$

$$\sigma_k^{(t+1)} = \frac{\sum_{m=1}^M \mathbf{d}_{mk} p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}) (1 + \gamma \ln p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}))}{\sum_{m=1}^M p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}) (1 + \gamma \ln p(\mathbf{z}_k | \mathbf{x}_m, \mu^{(t)}, \sigma^{(t)}))} \quad (87)$$

where $\mathbf{d}_{mk} = (\mathbf{x}_m - \mu_k)(\mathbf{x}_m - \mu_k)^T$

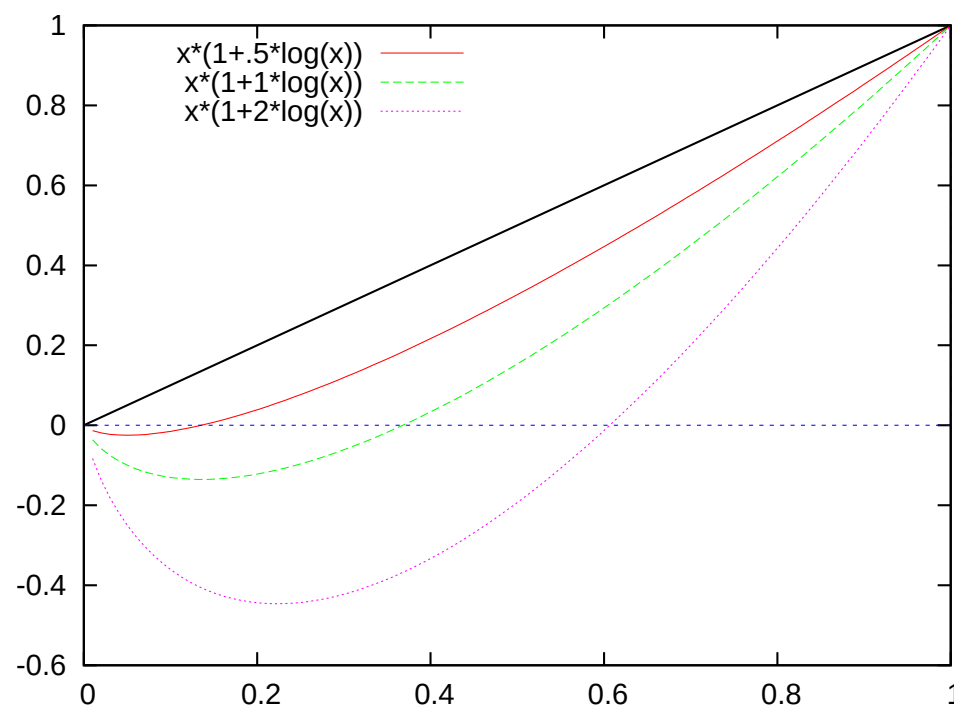
EM algorithm: Regularization

Comparing (78)-(78) and (84) just for parameter π_k :

$$\pi_k^{(t+1)} = \frac{\sum_{m=1}^M p(\mathbf{z}_k | \mathbf{x}_m, \pi_k^{(t)}, \mu_k, \sigma_k)}{\sum_{m=1}^M \sum_{k=1}^K p(\mathbf{z}_k | \mathbf{x}_m, \pi_k^{(t)}, \mu_k, \sigma_k)} = \frac{\sum_{m=1}^M p(\mathbf{z}_k | \mathbf{x}_m, \pi_k^{(t)}, \mu_k, \sigma_k)}{M}$$

$$\pi_k^{(t+1)} = \frac{\sum_{m=1}^M p(\mathbf{z}_k | \mathbf{x}_m, \pi_k^{(t)}, \mu_k, \sigma_k) (1 + \gamma \ln p(\mathbf{z}_k | \mathbf{x}_m, \pi_k^{(t)}, \mu_k, \sigma_k))}{\sum_{m=1}^M \sum_{k=1}^K p(\mathbf{z}_k | \mathbf{x}_m, \pi_k^{(t)}, \mu_k, \sigma_k) (1 + \gamma \ln p(\mathbf{z}_k | \mathbf{x}_m, \pi_k^{(t)}, \mu_k, \sigma_k))}$$

Contribution of each sample in the numerator



EM algorithm: Regularization

No regularization

$$\theta^{(0)} = (\pi_1 = 0.700, \pi_2 = 0.100, \pi_3 = 0.200)$$

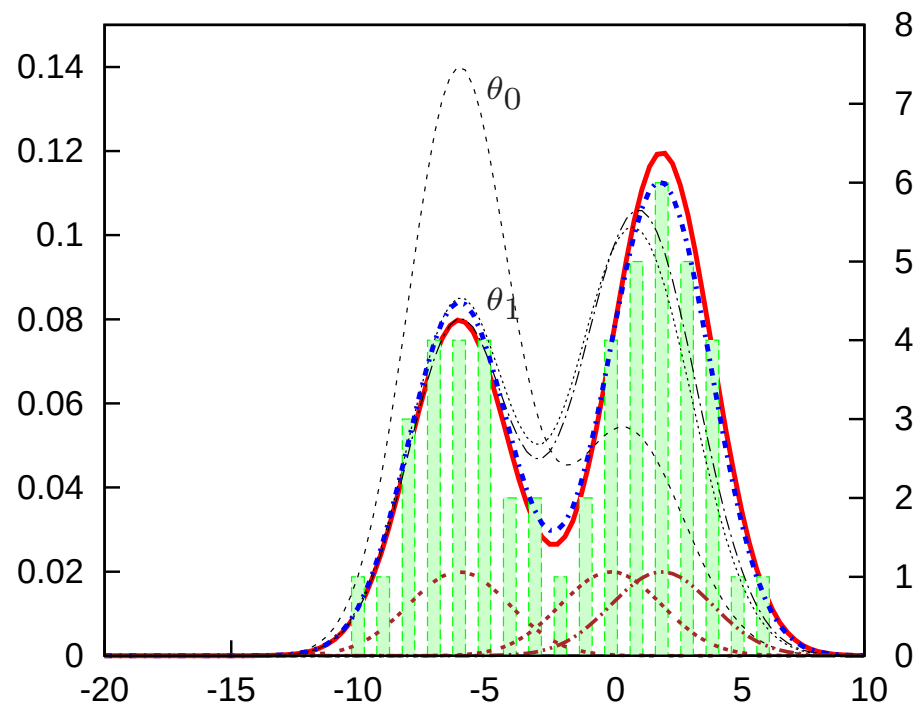
$$\theta^{(1)} = (\pi_1 = 0.423, \pi_2 = 0.259, \pi_3 = 0.318)$$

$$\theta^{(2)} = (\pi_1 = 0.399, \pi_2 = 0.320, \pi_3 = 0.281)$$

$$\theta^{(3)} = (\pi_1 = 0.402, \pi_2 = 0.391, \pi_3 = 0.207)$$

... (26 iterations later)

$$\theta^{(26)} = (\pi_1 = 0.421, \pi_2 = 0.543, \pi_3 = 0.036)$$



Regularization $\gamma = .1$

$$\theta^{(0)} = (\pi_1 = 0.700, \pi_2 = 0.100, \pi_3 = 0.200)$$

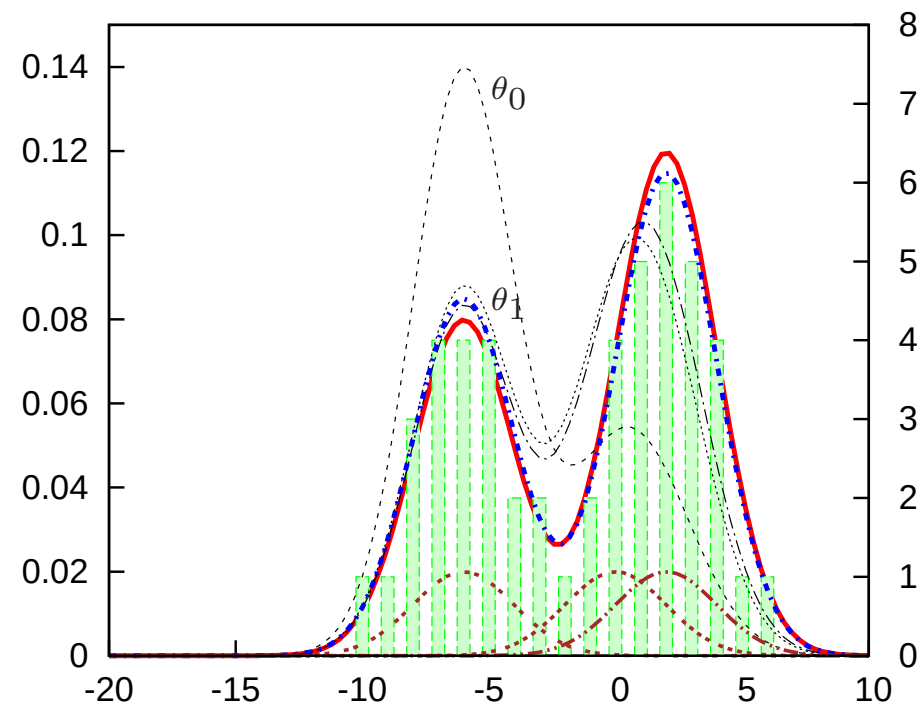
$$\theta^{(1)} = (\pi_1 = 0.438, \pi_2 = 0.251, \pi_3 = 0.311)$$

$$\theta^{(2)} = (\pi_1 = 0.415, \pi_2 = 0.315, \pi_3 = 0.270)$$

$$\theta^{(3)} = (\pi_1 = 0.418, \pi_2 = 0.385, \pi_3 = 0.197)$$

... (13 iterations later)

$$\theta^{(13)} = (\pi_1 = 0.426, \pi_2 = 0.574, \pi_3 = 0.000)$$



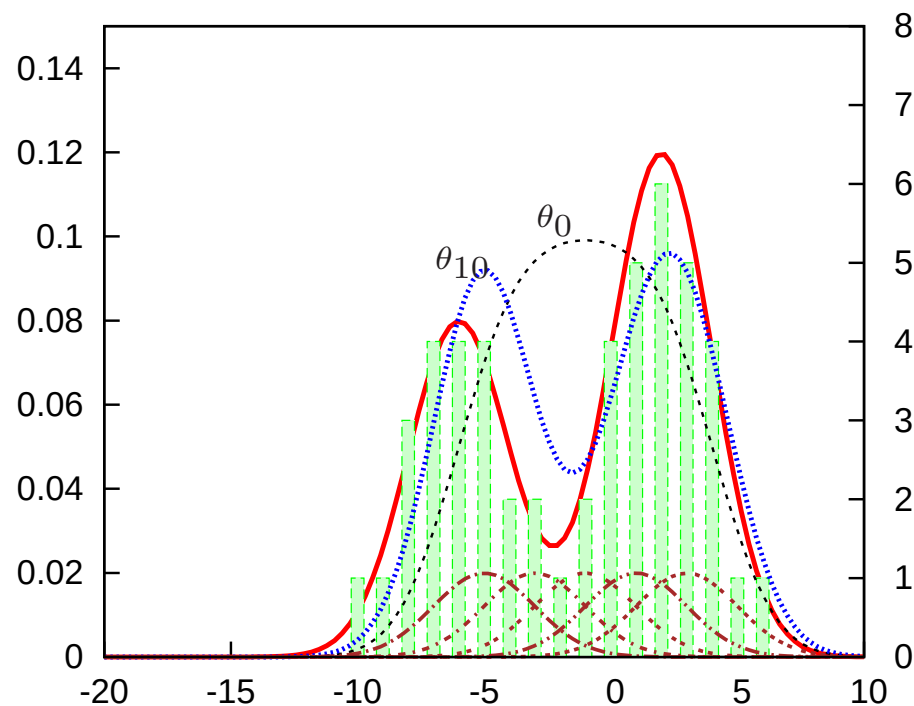
EM algorithm: Regularization

Regularization $\gamma = .1$

$$\theta^{(0)} = (\pi_1=0.200, \pi_2=0.200, \pi_3=0.200, \pi_4=0.200, \pi_5=0.200;$$

$$\mu_1 = -5.0, \mu_2 = -3.0, \mu_3 = -1.0, \mu_4 = 1.0, \mu_5 = 3.0)$$

$$\theta^{(10)} = (\pi_1=0.459, \pi_2=0.000, \pi_3=0.000, \pi_4=0.204, \pi_5=0.336)$$

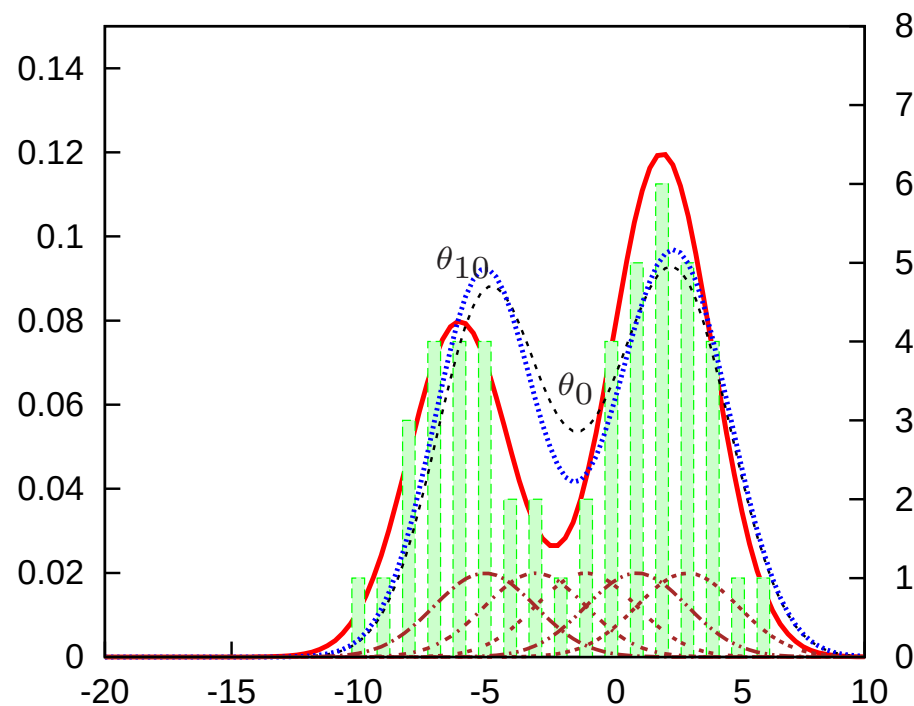


Regularization $\gamma = .1$

$$\theta^{(0)} = (\pi_1=0.400, \pi_2=0.050, \pi_3=0.050, \pi_4=0.150, \pi_5=0.350;$$

$$\mu_1 = -5.0, \mu_2 = -3.0, \mu_3 = -1.0, \mu_4 = 1.0, \mu_5 = 3.0)$$

$$\theta^{(10)} = (\pi_1=0.460, \pi_2=0.000, \pi_3=0.000, \pi_4=0.194, \pi_5=0.346)$$



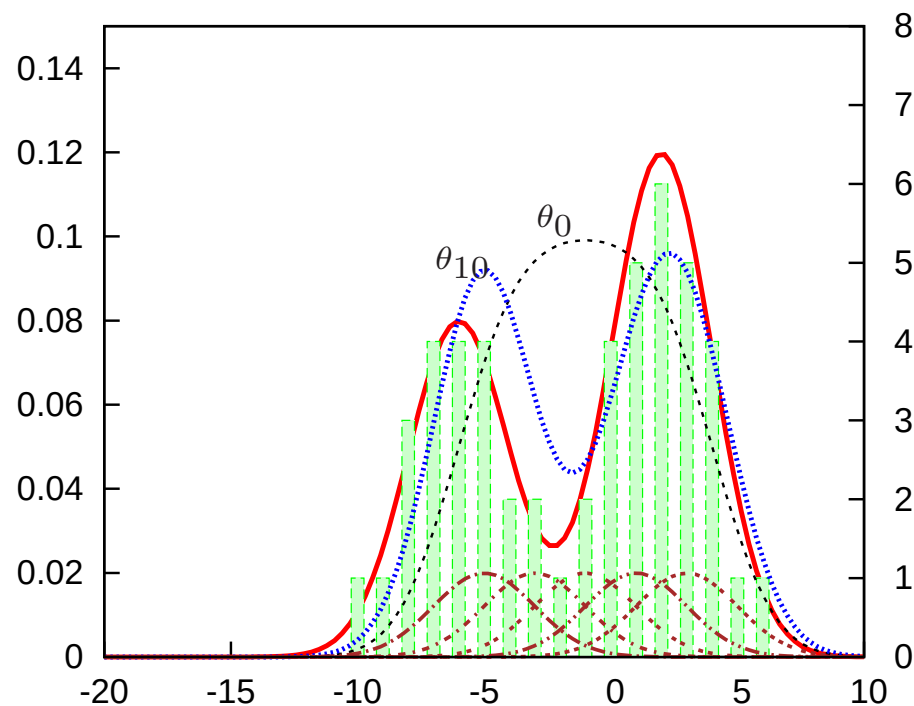
EM algorithm: Regularization

Regularization $\gamma = .1$

$$\theta^{(0)} = (\pi_1=0.200, \pi_2=0.200, \pi_3=0.200, \pi_4=0.200, \pi_5=0.200;$$

$$\mu_1 = -5.0, \mu_2 = -3.0, \mu_3 = -1.0, \mu_4 = 1.0, \mu_5 = 3.0)$$

$$\theta^{(10)} = (\pi_1=0.459, \pi_2=0.000, \pi_3=0.000, \pi_4=0.204, \pi_5=0.336)$$

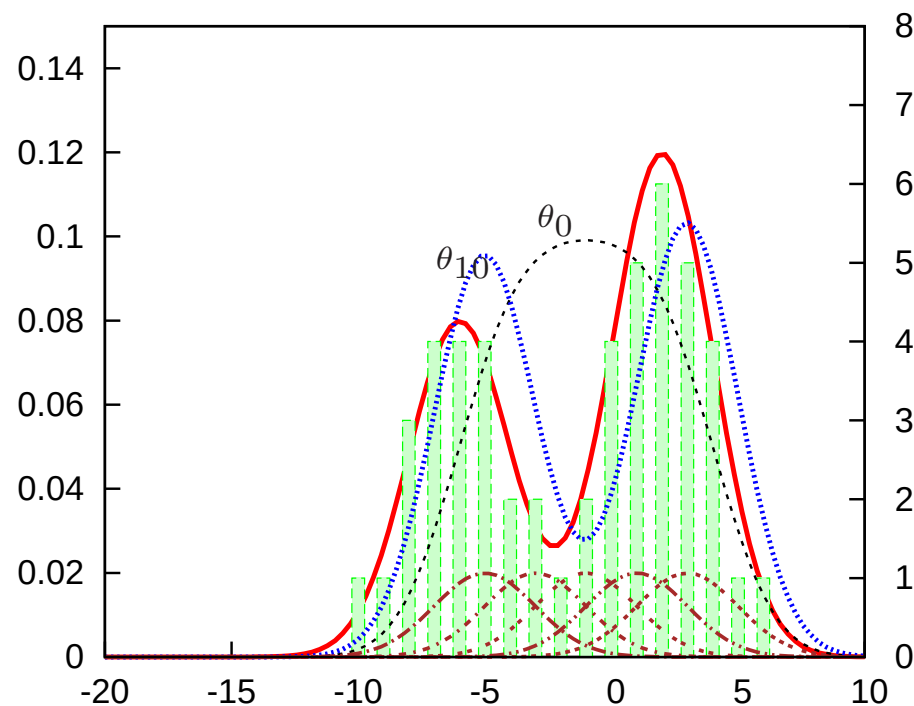


Regularization $\gamma = 0.3$

$$\theta^{(0)} = (\pi_1=0.200, \pi_2=0.200, \pi_3=0.200, \pi_4=0.200, \pi_5=0.200;$$

$$\mu_1 = -5.0, \mu_2 = -3.0, \mu_3 = -1.0, \mu_4 = 1.0, \mu_5 = 3.0)$$

$$\theta^{(10)} = (\pi_1=0.479, \pi_2=0.000, \pi_3=0.000, \pi_4=0.009, \pi_5=0.512)$$



EM algorithm: Regularization

2D. EM with regularization. 10 initial clusters.

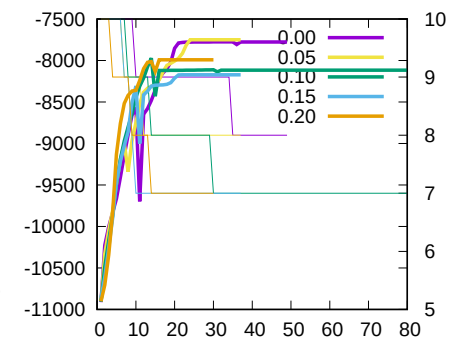
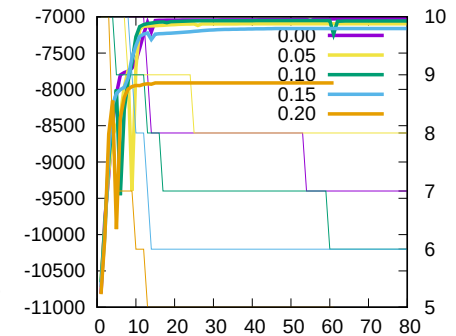
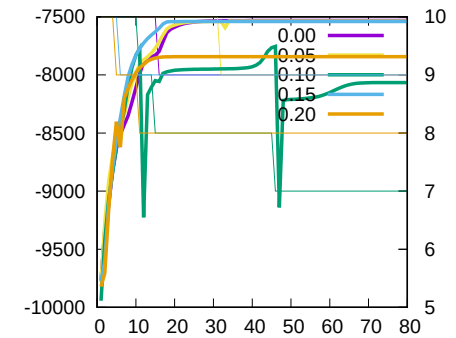
$\gamma = 0.00$

$\gamma = 0.05$

$\gamma = 0.10$

$\gamma = 0.15$

$\gamma = 0.20$



EM algorithm: Regularization

2D. EM with regularization. 20 initial clusters.

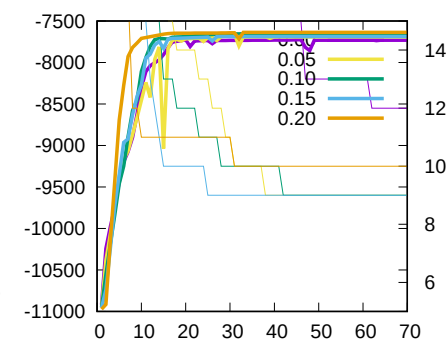
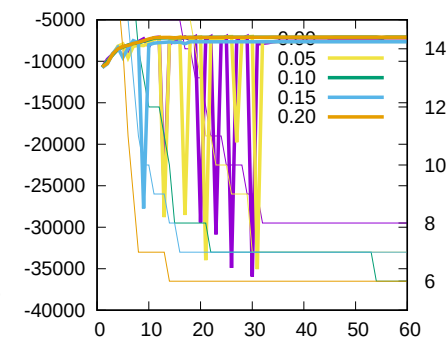
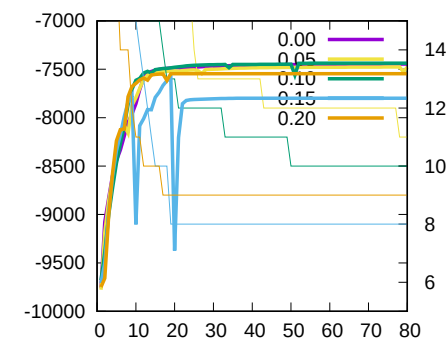
$\gamma = 0.00$

$\gamma = 0.05$

$\gamma = 0.10$

$\gamma = 0.15$

$\gamma = 0.20$



EM algorithm: Regularization

Pending issues:

- Decide the value of γ : a validation set, ...
- Criterion to remove a gaussian
- Initialize the algorithm: c-means, ...
- Dimensionality

Exercises

Exercise (*):** Reproduce an example similar to the unidimensional example with three unidimensional distributions, with known mean and equal and known variance where only π_1, π_2, π_3 are unknown. Obtain the corresponding plots.

Exercise (*):** Reproduce an example similar to the unidimensional example with two bidimensional distributions, with known mean vector and equal and known co-variance matrix where only π_1, π_2, π_3 are unknown. Obtain the corresponding plots.

Exercise (**):** Reproduce an example similar to the unidimensional example with two unidimensional distributions, with equal and known variance where the mean of each distribution and π_1, π_2 are unknown. Obtain the corresponding plots.

Exercise (**):** Reproduce an example similar to the unidimensional example with two unidimensional distributions, with equal and known mean where the variances and π_1, π_2 are unknown. Obtain the corresponding plots.

Exercise (**):** Reproduce an example similar to the bi-dimensional example, but the initialization has to be performed with the c-means algorithm. Obtain the corresponding plots.

Exercise (***):** Reproduce an example similar to the bi-dimensional example, but the initialization has to be performed with the c-means algorithm. Apply the algorithm to a real classification task.

Exercise (***):** Apply the described regularization method to other type of models (e.g. HMM)

EM algorithm: Regularization

Regularization applied to grammatical models (PFA)

From (80), let $\mathbf{W}$ a variable associated to the strings of a PFA $\mathcal{A}$ and Θ a variable associated to a state sequence, and ϑ the parameters of the model:

$$\tilde{L}(\mathbf{W}|\vartheta) = L(\mathbf{X}|\vartheta) + \gamma I(\mathbf{W}, \Theta|\vartheta) = L(\mathbf{X}|\theta) + \gamma (H(\Theta) - H(\Theta|\mathbf{W}))$$

Using (27), then:

$$\tilde{L}(\mathbf{W}|\vartheta) = L(\mathbf{W}|\vartheta) + \gamma H(\mathbf{W}|\vartheta) \quad (88)$$

Another point of view: Since $\mathbf{W}$ is a deterministic function of Θ (the contrary is not true) then the mutual information coincides with $H(\mathbf{W}|\vartheta)$ (see plot in page 11)

Estimation expression: (Hypothesis!)

$$\bar{P}(i, v, j) = \frac{\sum_w \frac{1}{p_{\mathcal{A}}(w)} \sum_{\theta_w} N((i, v, j), \theta_w) p_{\mathcal{A}}(w, \theta_w) (1 + \gamma \log p_{\mathcal{A}}(w))}{\sum_w \frac{1}{p_{\mathcal{A}}(w)} \sum_{\theta_w} N(i, \theta_w) p_{\mathcal{A}}(x, \theta_x) (1 + \gamma \log p_{\mathcal{A}}(w))} \quad (89)$$

Exercise (**):** Proof if the previous expression is true or false!

Index

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2. Entropy measures for grammatical models
3. Maximum entropy models
4. Regularized EM algorithm
5. **Discriminative training criterion**
6. Semi-supervised learning
7. Active learning

Discriminative training criterion

H-criterion based training [Gopalakrishnan 88]. Let a, b, c constants with $a > 0$:

$$H_{a,b,c}(\theta) = -\frac{1}{n} \sum_{i=1}^n \log p_{\theta}^a(\mathbf{x}_i, \mathbf{z}_i) p_{\theta}^b(\mathbf{x}_i) p_{\theta}^c(\mathbf{z}_i) \quad (90)$$

Observe the close relation between the previous expression and the Mutual Information (9)

Maximal Mutual Information

$$F(\theta) = \frac{1}{n} \sum_{i=1}^n \log \frac{p_{\theta}(\mathbf{x}_i | \mathbf{z}_i)}{p_{\theta}(\mathbf{x}_i)} \quad (91)$$

$$= \frac{1}{n} \sum_{i=1}^n \log \frac{p_{\theta}(\mathbf{x}_i, \mathbf{z}_i)}{p_{\theta}(\mathbf{x}_i) p_{\theta}(\mathbf{z}_i)} \quad (92)$$

$$\approx H_{\theta}(X) + H_{\theta}(Z) - H_{\theta}(X, Z) \quad (93)$$

Thus $F(\theta) = H_{1,-1,-1}(\theta)$ in expression (90)

The join distribution $p(\mathbf{x}, \mathbf{z})$ in expression (9) is “replaced” in (92) by an empirical distribution $\tilde{p}(\mathbf{x}, \mathbf{z})$

Discriminative training criterion

MMI Criterion. The aim is to estimate the parameters in such a way as to (approximately) reduce the error rate on the training data.

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^n \log \frac{p_{\theta}(\mathbf{x}_i | \mathbf{z}_i)}{p_{\theta}(\mathbf{x}_i)} = \arg \max_{\theta} \sum_{i=1}^n \log \frac{p_{\theta}(\mathbf{x}_i | \mathbf{z}_i)}{\sum_j p_{\theta}(\mathbf{x}_i | \mathbf{z}_j) p_{\theta}(\mathbf{z}_j)} \quad (94)$$

MMI Criterion - Alternative: Use $F(\theta) = H_{1,-h,0}(\theta)$ in expression (90) with $0 \leq h \leq 1$. Value h aims to establish the degree that competing derivation discriminate against the reference derivation. The optimization of the H-criterion aims to maximize the numerator while decreasing the denominator.¹

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^n \log \frac{\sum_{j'} (p_{\theta}(\mathbf{x}_i, \mathbf{z}_{j'}))^{\nu}}{\left(\sum_j (p_{\theta}(\mathbf{x}_i, \mathbf{z}_j))^{\nu} \right)^h} \quad (95)$$

with $\nu > 0$

¹ **Important:** The sum in the denominator has to include the terms in the numerator.

Discriminative training criterion

Problem: expression (95) has a rational function

Solution: Growth transformation for rational functions [Gopalakrishnan 1991]

Example: SCFG with parameters ϑ [Maca 2021]

$$\bar{P}(A \rightarrow \alpha) = \frac{\sum_x \frac{1}{p_{\vartheta}^{\nu}(x, \Delta')} \sum_{t'_x \in \Delta'} N(A \rightarrow \alpha, t'_x) p_{\vartheta}^{\nu}(t'_x) - h \sum_x \frac{1}{p_{\vartheta}^{\nu}(x, \Delta)} \sum_{t_x \in \Delta} N(A \rightarrow \alpha, t_x) p_{\vartheta}^{\nu}(t_x) + p(A \rightarrow \alpha) C^*(\vartheta)}{\sum_x \frac{1}{p_{\vartheta}^{\nu}(x, \Delta')} \sum_{t'_x \in \Delta'} N(A, t'_x) p_{\vartheta}^{\nu}(t'_x) - h \sum_x \frac{1}{p_{\vartheta}^{\nu}(x, \Delta)} \sum_{t_x \in \Delta} N(A \rightarrow \alpha, t_x) p_{\vartheta}^{\nu}(t_x) + C^*(\vartheta)} \quad (96)$$

where $\Delta' \subseteq \Delta$ are subset of derivation of string x and

$$C^*(\vartheta) = \max \left(\max_{p(A \rightarrow \alpha)} \left(- \frac{\sum_x \frac{1}{p_{\vartheta}^{\nu}(x, \Delta')} \sum_{t'_x \in \Delta'} N(A \rightarrow \alpha, t'_x) p_{\vartheta}^{\nu}(t'_x) - h \sum_x \frac{1}{p_{\vartheta}^{\nu}(x, \Delta)} \sum_{t_x \in \Delta} N(A \rightarrow \alpha, t_x) p_{\vartheta}^{\nu}(t_x)}{p(A \rightarrow \alpha)} \right), 0 \right) \quad (97)$$

Discriminative training criterion

MMI Criterion for CHMM [Povey 2001]. For M observation sequences $\{O_1, O_2, \dots, O_M\}$ choose λ that maximizes:

$$\mathcal{F}_{\text{mmi}}(\lambda) = \frac{1}{M} \sum_{i=1}^M \log p_{\lambda}(w_i^{\text{ref}} | O_i) = \frac{1}{M} \sum_{i=1}^M \log \frac{p_{\lambda}(O_i | w_i^{\text{ref}}) P(w_i^{\text{ref}})}{\sum_{\hat{w}} p_{\lambda}(O_i | \hat{w}) P(\hat{w})} \quad (98)$$

where w is a composite model corresponding to the word sequence represented in O . Note that summation for $\hat{w}$ is not feasible. In practice this is restricted to the set of confusable hypotheses (see also [Schlüter 2001])

Discriminative training criterion

MPE Criterion for CHMM [Povey 2002]. To minimize the function:

$$\mathcal{F}_{\text{mpe}}(\lambda) = \sum_{i=1}^M \sum_w p_{\lambda}(w|O_i) \mathcal{L}(w, w_i^{\text{ref}}) \quad (99)$$

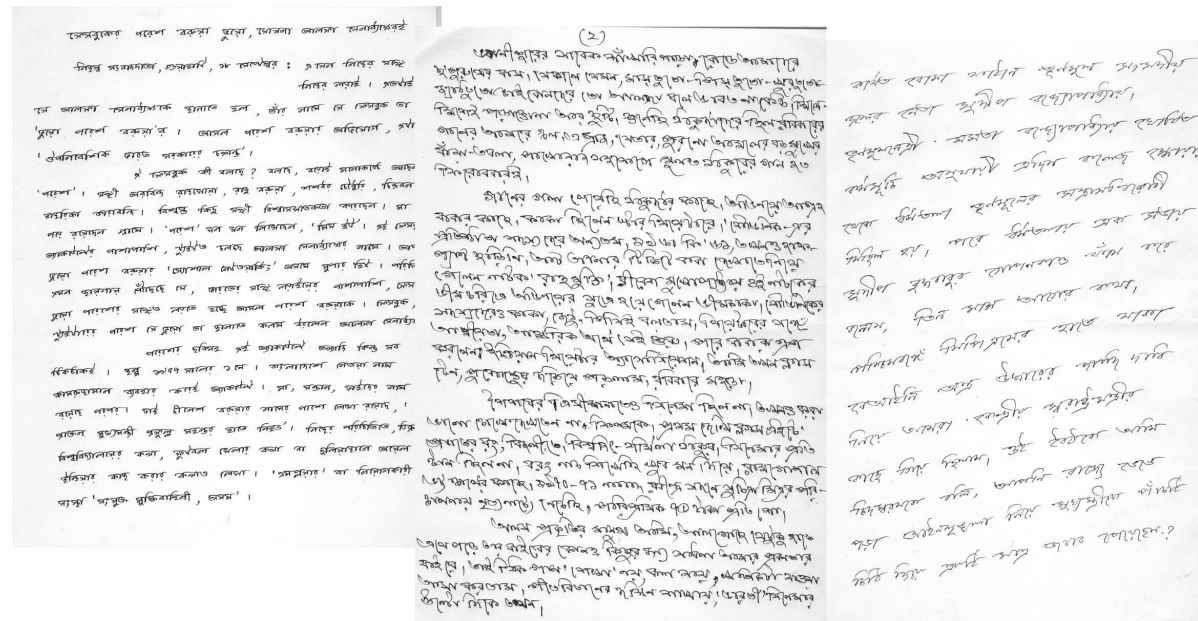
where $\mathcal{L}(w, w_i^{\text{ref}})$ is the “loss” between the hypothesis w and the reference w_i^{ref} . In MPE the “loss” is based on the Levenshtein edit distance between the phone sequences of the reference and the hypothesis (see HTK manual for details)

Discriminative training criterion

Experiments [Sánchez 2016]: Handwritten Text Recognition of Bengali using holistic segmentation-free Handwritten Text Recognition (HTR) technology

Characteristics of Bengali:

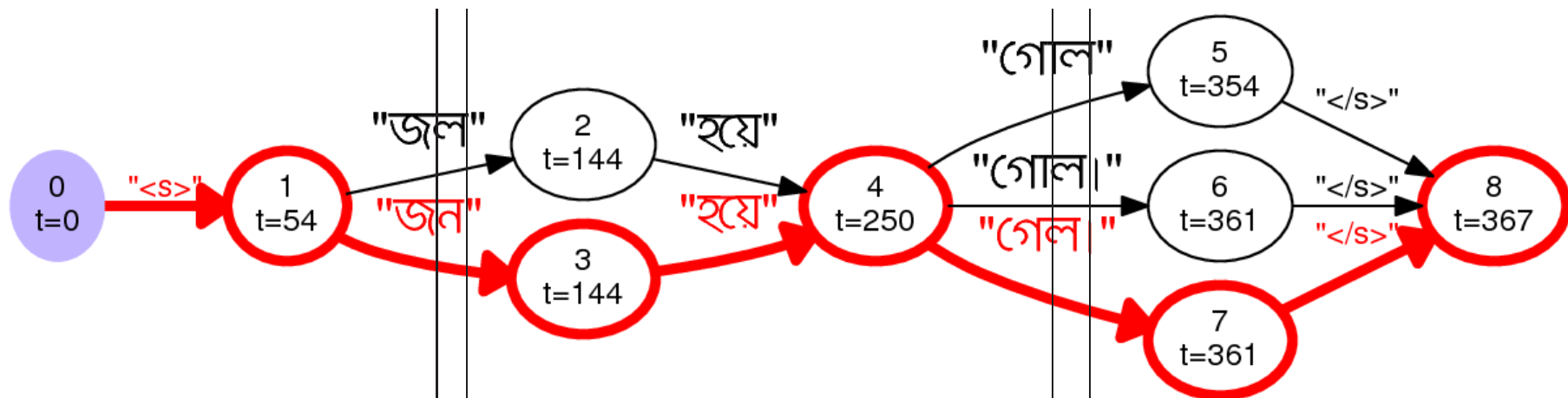
- About 300 different characters
- About 90 characters appears 3 or less times in the dataset



Discriminative training criterion

Training Criterion: First ML, and then MPE

- The numerator is the real transcript and it is exactly the log-likelihood
- The denominator includes the competitor transcripts (restricted to be the most probable transcripts according to a less restrictive LM: word-based LM or character-based LM)



Discriminative training criterion

Number of:	P0	Total
Pages	20	98
Lines	465	1,785
Run. words	4,407	15,694
Run. OOV	1,503	-
Lexicon	1,953	4,962
Char. set size	217	291
Run. Characters	5,148	42,712

- Cross-validation: 5 partitions
- About 30% OOV in average

- ▶ Feature extraction based on moments computed from a sliding window
- ▶ No special treatment for modifiers placed out of the main body: ூ, ூ, ...
Left to the LM model
- ▶ Two-stroked characters (ㄅ) modeled with two HMM. The lexical model take care of reversing the problem

Discriminative training criterion

Results. Recognition with word-based LM (2-gram)

	All Parts.		P0	
	WER	CER	WER	CER
Baseline	63.4	39.3	67.9	42.8
+ Gen. HMM	62.7	38.5	66.7	40.7
+ CV	49.7	31.8	52.2	32.8
+ WL-MPE	-	-	50.4	29.8
+ CL-MPE-1	-	-	43.8	25.8
+ CL-MPE-2	-	-	43.0	25.4

Problem: 88 characters out of approximately 290 appeared 3 times or less in the dataset

Solution: to substitute by a generic HMM

বেলায়	রোদ্দুর	পড়ে	আসে	;	গা	খুলে	বেড়ায়
বেলায়	রোদে তার	পরে	আসে	;	গা	বিকেল	বেলায়
বেলগাছে	রোদ্দুর	পড়ে	আসে	;	গা	বিকেল	বেলায়

Index

1. Entropy definitions
2. Entropy measures for grammatical models
3. Maximum entropy models
4. Regularized EM algorithm
5. Discriminative training criterion
6. **Semi-supervised learning**
7. Active learning

Semi-supervised learning

Problem: Classification

- Supervised training: both label y and object representation $\mathbf{x}$ are available
- Unsupervised training: only object representation $\mathbf{x}$ are available
- Generative models: $p(y, \mathbf{x})$
- Discriminative models: $p(y|\mathbf{x})$

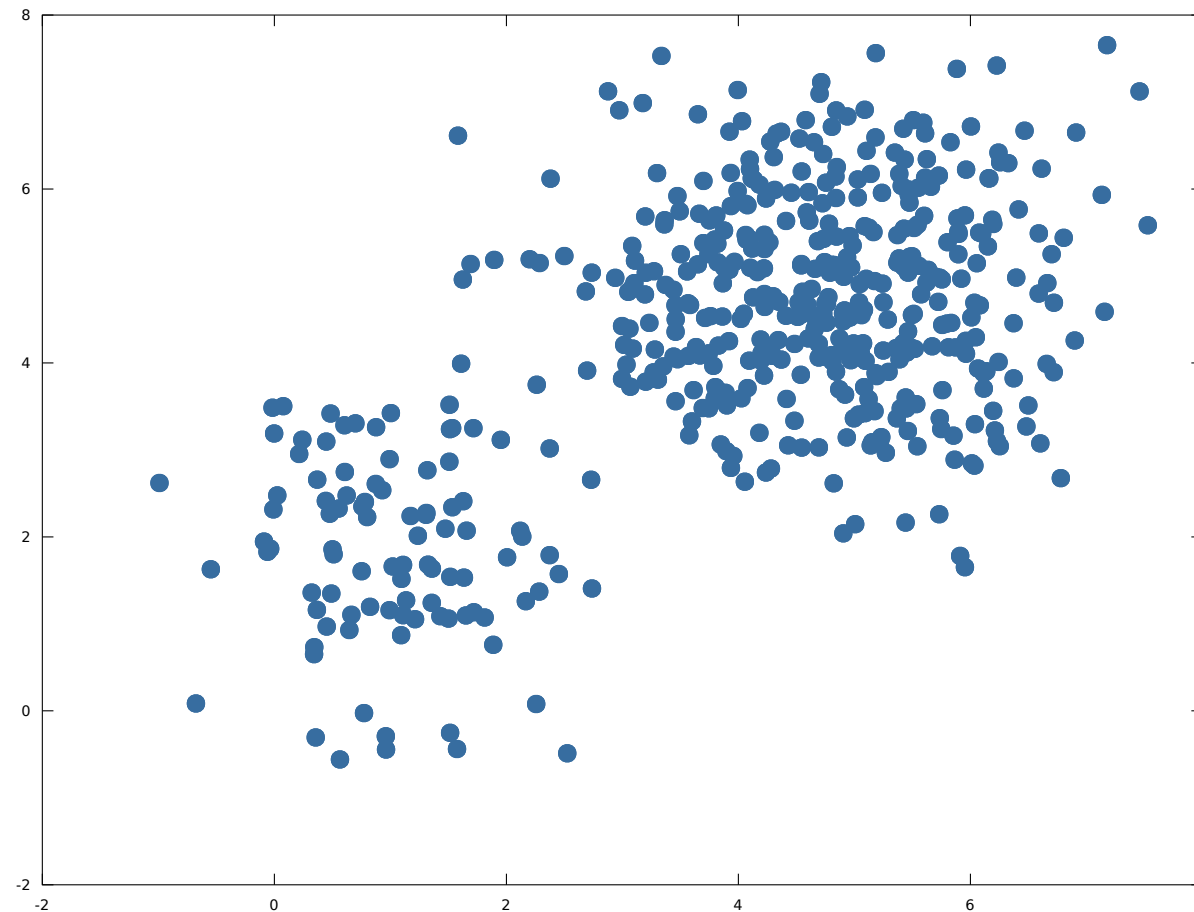
Problem: Unsupervised learning

- Regularized EM for gaussians?
- Regularized EM for gaussians with some help? Information about the class priors?

Semi-supervised learning

Intuition

Can we guess the two classes given the priors $p(c_1) = 0.19$ and $p(c_2) = 0.81$?



Semi-supervised learning

Unsupervised learning for generative models:

$$\arg \max_{\theta} \log p_{\theta}(x) = \arg \max_{\theta} \log \sum_y p_{\theta}(x|y) p(y)$$

Some preliminary papers [Ravi 2011]

For discriminative models: ?

Semi-supervised learning

Formal statement

Consider a labeled sample $(\mathbf{X}, \mathbf{Z})$ and a unlabeled sample $(\tilde{\mathbf{X}})$ (IID). Let us assume that we know the labels of $(\tilde{\mathbf{X}})$ and we are using a *generative* model.

From expression (71):

$$\Theta_{n+1} = \arg \max_{\theta} \left\{ \sum_{\mathbf{z}} \mathcal{P}(\mathbf{z}|\mathbf{X}, \theta_n) \ln \mathcal{P}(\mathbf{X}, \mathbf{z}|\theta) \right\}$$

If we assume that the unknown labels $\tilde{\mathbf{Z}}$ are given for $\tilde{\mathbf{X}}$, then the “complete-data” likelihood is:

$$\mathcal{P}(\mathbf{X}, \mathbf{Z}, \tilde{\mathbf{X}}, \tilde{\mathbf{Z}}) = \mathcal{P}(\mathbf{X}, \mathbf{Z}) \mathcal{P}(\tilde{\mathbf{X}}, \tilde{\mathbf{Z}}) \quad (100)$$

Objective function

$$\mathcal{P}(\mathbf{X}, \mathbf{Z}, \tilde{\mathbf{X}}) = \sum_{\tilde{\mathbf{z}}} \mathcal{P}(\mathbf{X}, \mathbf{Z}, \tilde{\mathbf{X}}, \tilde{\mathbf{z}}) \quad (101)$$

Semi-supervised learning

$$\begin{aligned}
Q(\Theta \mid \Theta_n) &= \mathbb{E}_{\tilde{\mathbf{Z}} \mid \mathbf{X}, \mathbf{Z}, \tilde{\mathbf{X}}, \Theta_n} \ln \mathcal{P}(\mathbf{X}, \mathbf{Z}, \tilde{\mathbf{X}}, \tilde{\mathbf{Z}}) \\
&= \sum_{\tilde{\mathbf{Z}}} \mathcal{P}(\tilde{\mathbf{Z}} \mid \mathbf{X}, \mathbf{Z}, \tilde{\mathbf{X}}, \Theta_n) \left(\ln \mathcal{P}(\mathbf{X}, \mathbf{Z} \mid \Theta) + \ln \mathcal{P}(\tilde{\mathbf{X}}, \tilde{\mathbf{Z}} \mid \Theta) \right) \\
&= \sum_{i=1}^n \ln \mathcal{P}(\mathbf{x}^{(i)}, \mathbf{z}^{(i)} \mid \Theta) \sum_{\tilde{\mathbf{Z}}} \mathcal{P}(\tilde{\mathbf{Z}} \mid \mathbf{X}, \mathbf{Z}, \tilde{\mathbf{X}}, \Theta_n) \\
&\quad + \sum_{j=1}^m \sum_{\tilde{\mathbf{Z}}} \mathcal{P}(\tilde{\mathbf{Z}} \mid \mathbf{X}, \mathbf{Z}, \tilde{\mathbf{X}}, \Theta_n) \ln \mathcal{P}(\tilde{\mathbf{x}}^{(j)}, \tilde{\mathbf{z}}^{(j)} \mid \Theta) \\
&= \sum_{i=1}^n \ln \mathcal{P}(\mathbf{x}^{(i)}, \mathbf{z}^{(i)} \mid \Theta) + \sum_{j=1}^m \sum_{\tilde{\mathbf{Z}}} \mathcal{P}(\tilde{\mathbf{Z}} \mid \tilde{\mathbf{X}}, \Theta_n) \ln \mathcal{P}(\tilde{\mathbf{x}}^{(j)}, \tilde{\mathbf{z}}^{(j)} \mid \Theta) \\
&= \sum_{i=1}^n \ln \mathcal{P}(\mathbf{x}^{(i)}, \mathbf{z}^{(i)} \mid \Theta) + \sum_{j=1}^m \sum_{\tilde{\mathbf{Z}}} \left(\prod_{j'=1}^m \mathcal{P}(\tilde{\mathbf{Z}} \mid \tilde{\mathbf{x}}^{(j')}, \Theta_n) \right) \ln \mathcal{P}(\tilde{\mathbf{x}}^{(j)}, \tilde{\mathbf{z}}^{(j)} \mid \Theta) \\
&= \sum_{i=1}^n \ln \mathcal{P}(\mathbf{x}^{(i)}, \mathbf{z}^{(i)} \mid \Theta) + \sum_{j=1}^m \sum_{\tilde{\mathbf{z}}^{(j)}} \mathcal{P}(\tilde{\mathbf{z}}^{(j)} \mid \tilde{\mathbf{x}}^{(j)}, \Theta_n) \ln \mathcal{P}(\tilde{\mathbf{x}}^{(j)}, \tilde{\mathbf{z}}^{(j)} \mid \Theta) \quad (102)
\end{aligned}$$

Semi-supervised learning

Note that:

$$\mathcal{P}(\tilde{\mathbf{z}}^{(j)} \mid \tilde{\mathbf{x}}^{(j)}, \Theta_n) \quad (103)$$

are the current estimates for the probabilities of each of the labels of the unlabeled examples. These can be computed as:

$$\mathcal{P}(\tilde{\mathbf{z}}^{(j)} \mid \tilde{\mathbf{x}}^{(j)}, \Theta_n) = \frac{\mathcal{P}(\tilde{\mathbf{z}}^{(j)}, \tilde{\mathbf{x}}^{(j)} \mid \Theta_n)}{\sum_{\tilde{\mathbf{z}}} \mathcal{P}(\tilde{\mathbf{z}}, \tilde{\mathbf{x}} \mid \Theta_n)} \quad (104)$$

The extension to multi-class is straightforward.

Note that:

$$\mathcal{P}(\tilde{\mathbf{z}}^{(j)}, \tilde{\mathbf{x}}^{(j)} \mid \Theta_n) = \mathcal{P}(\tilde{\mathbf{z}}^{(j)} \mid \Theta_n) \mathcal{P}(\tilde{\mathbf{x}}^{(j)} \mid \tilde{\mathbf{z}}^{(j)}, \Theta_n)$$

For HTR, a language model can be used and only the “best” decoding can be used:

$$\hat{\mathbf{z}} = \arg \max_z \mathcal{P}_{\text{LM}}(\mathbf{z}) \mathcal{P}(\tilde{\mathbf{x}}^{(j)} \mid \mathbf{z})$$

Semi-supervised learning

Main idea in semi-supervised learning

1. Obtain initial models trained in a supervised way, with a small amount of annotated data
2. Use the current models for obtaining new “labeled” data from unannotated data
3. Use expression (104) to perform supervised model training
4. Repeat steps 2 and 3 until converge

Semi-supervised learning

Main idea in semi-supervised learning

1. Obtain initial models trained in a supervised way, with a small amount of annotated data
2. Use the current models with a **very good prior** for obtaining new data from unannotated data together with confidence measures
3. Use the more confident data (and the corresponding labels) from step 2 (and supervised samples from step 1) to perform supervised model training
4. Repeat steps 2 and 3 until converge

Semi-supervised learning

2D gaussians. 10 clusters. Confidence $> 50\%$

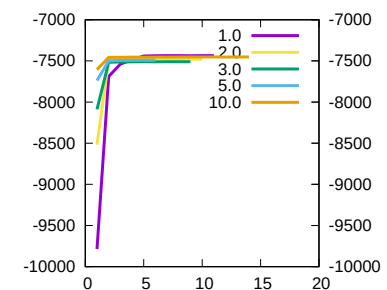
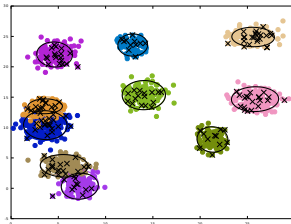
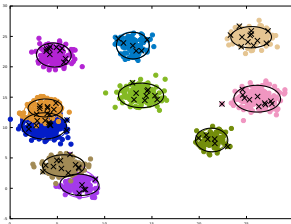
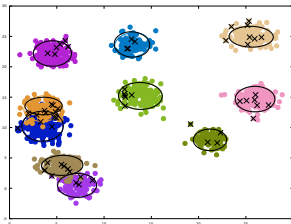
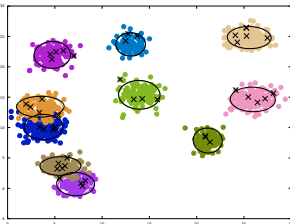
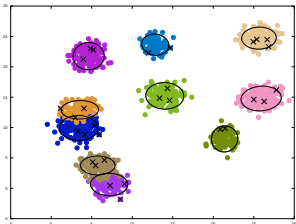
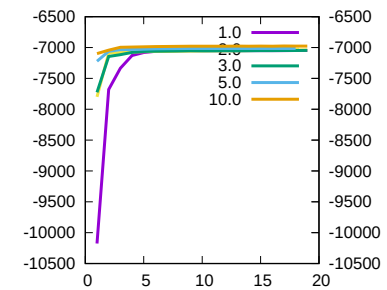
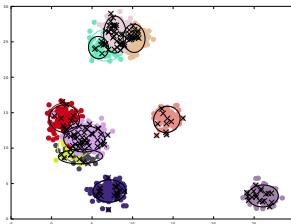
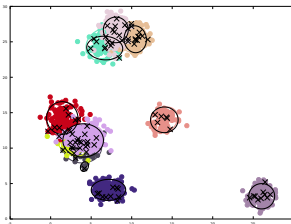
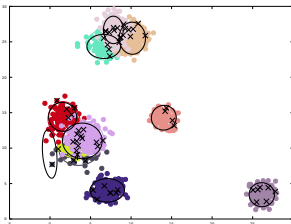
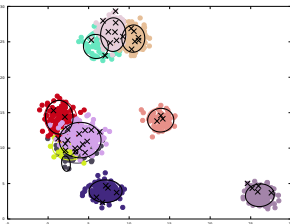
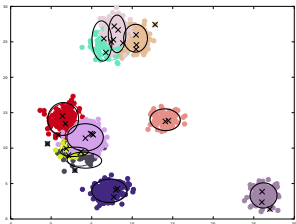
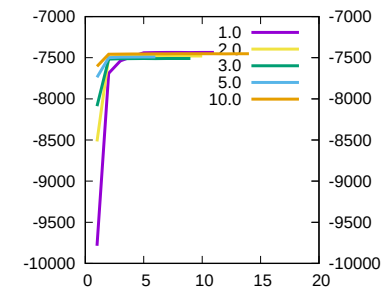
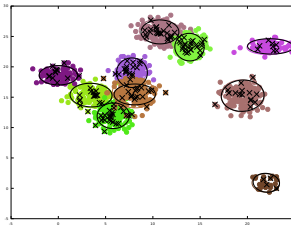
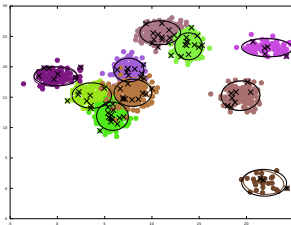
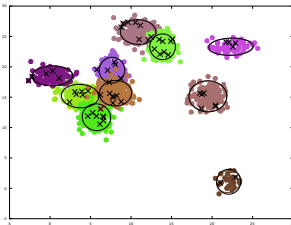
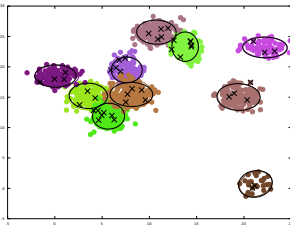
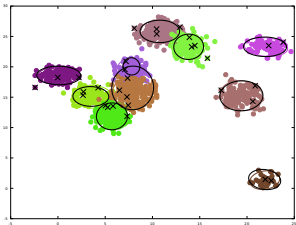
%= 1.0

%= 2.0

%= 3.0

%= 5.0

%= 10.0



Semi-supervised learning

Exercise (*):** Choose a data set and perform a classification experiment in a similar way to the previous slide. Plot the error rate as a function of the number of samples initially annotated.

Semi-supervised learning

Problem: Semi-supervised learning for Handwritten Text Recognition

Approach

$$\hat{\mathbf{w}} = \arg \max_{\mathbf{w}} P(\mathbf{w} \mid \mathbf{x}) = \arg \max_{\mathbf{w}} P(\mathbf{x} \mid \mathbf{w})P(\mathbf{w})$$

$P(\mathbf{w})$: language model $\longleftarrow n$ -grams

$P(\mathbf{x} \mid \mathbf{w})$: optical model $\longleftarrow$ CRNN-HMM

Training

- $P(\mathbf{w})$: n -grams training from plain text $\longleftarrow$ **easy and cheap**
- $P(\mathbf{x} \mid \mathbf{w})$: CRNN-HMM training from pairs
(line image, line transcript) $\longleftarrow$ **PROBLEM**

Semi-supervised learning

Training of $P(\mathbf{x} \mid \mathbf{w})$

Obtaining pairs (line image, line transcript):

- Supervised: **GT preparation is expensive**
- Unsupervised: **TO BE RESEARCHED**
- Semi-supervised: **Requires a lot of computation**

V. Frinken, M. Baumgartner, A. Fischer, and H. Bunke, “Semi-supervised learning for cursive handwriting recognition using keyword spotting”, ICFHR, 2012.

Semi-supervised learning

Main idea

1. Obtain initial models trained in a supervised way, with a small amount of transcribed line images
2. Use the current HMM and a **very good LM** for obtaining transcripts automatically from a large amount of untranscribed lines together with confidence measures (at character, word and/or sentence level)
(Remark: note that transcripts obtained automatically may have errors)
3. Use the more confident transcripts (and the corresponding images) from step 2 (and supervised samples from step 1) to perform model training with forced alignment
4. Repeat steps 2 and 3 until converge

Semi-supervised learning

Experiments with Benthan dataset [Sánchez, 2019]

- Training: 400 pages, 10,613 lines, 99K words
- Test: 33 pages, 860 lines, 7.8K words

Optical models: Semi-supervised learning of CRNN trained with Laia

Character-based 7-gram language model trained with:

- **Scenario 1. Supervised.** ICFHR 2014 training (Baseline)
- **Scenario 2. Semi-supervised.**
 - **Scenario 2.1.** Task dependent data: ICFHR 2014 training (99K words)
 - **Scenario 2.2.** Task independent data: Eighteenth Century Collections Online (ECCO)² database (2.8M words)

² <http://www.textcreationpartnership.org/tcp-ecco/>

Semi-supervised learning

Scenario 2. Semi-supervised.

- Start with 150 lines ($\approx$ 5 pages) for supervised training and 30 lines ($\approx$ 1 page) for development randomly selected $\rightarrow$ **Labeled data**
- Iterate until convergence:
 1. Train the optical models from scratch (CRNN training) with labeled data
 2. Transcribe the whole untranscribed training dataset
 3. Substitute characters with a confidence below 99% by a garbage character
 4. Retain sentences which transcript has a confidence above 95% and with less than 10% of incorrect characters for the next iteration. Keep these sentences of all iterations
 5. Sort and select the sentences according to the confidence score from this and previous iterations $\rightarrow$ **New labeled data maybe with errors**
 6. 10% is used for development (prioritize sentences without errors)

Semi-supervised learning

Scenario 2. Semi-supervised. Examples.

0. Application of the profit of the measure towards the

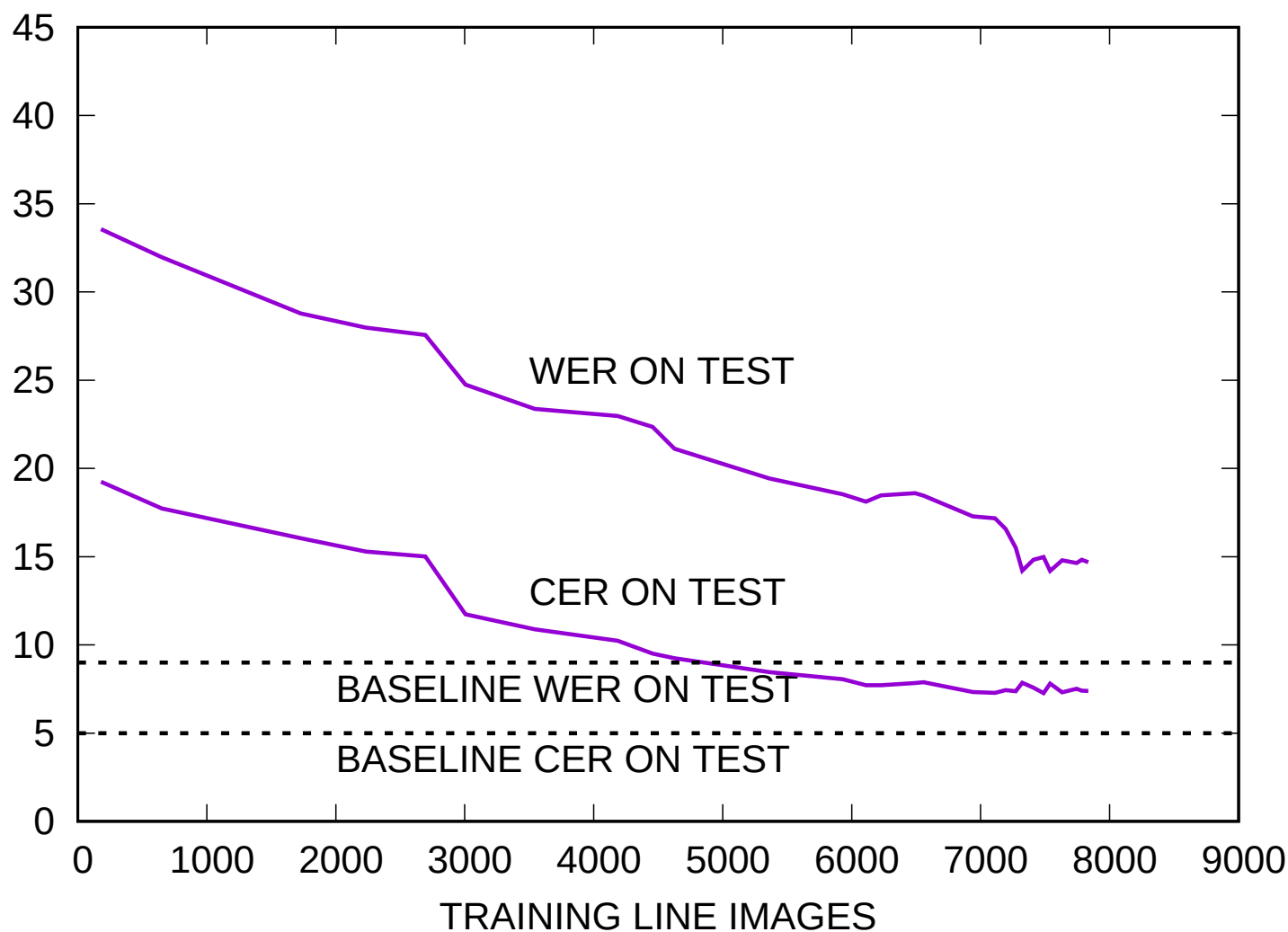
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0.92	1	vation by ether with the consideration of any
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0.97	0	vation by other with the consideration of any
1.00	0	ration together with the consideration of any

Semi-supervised learning

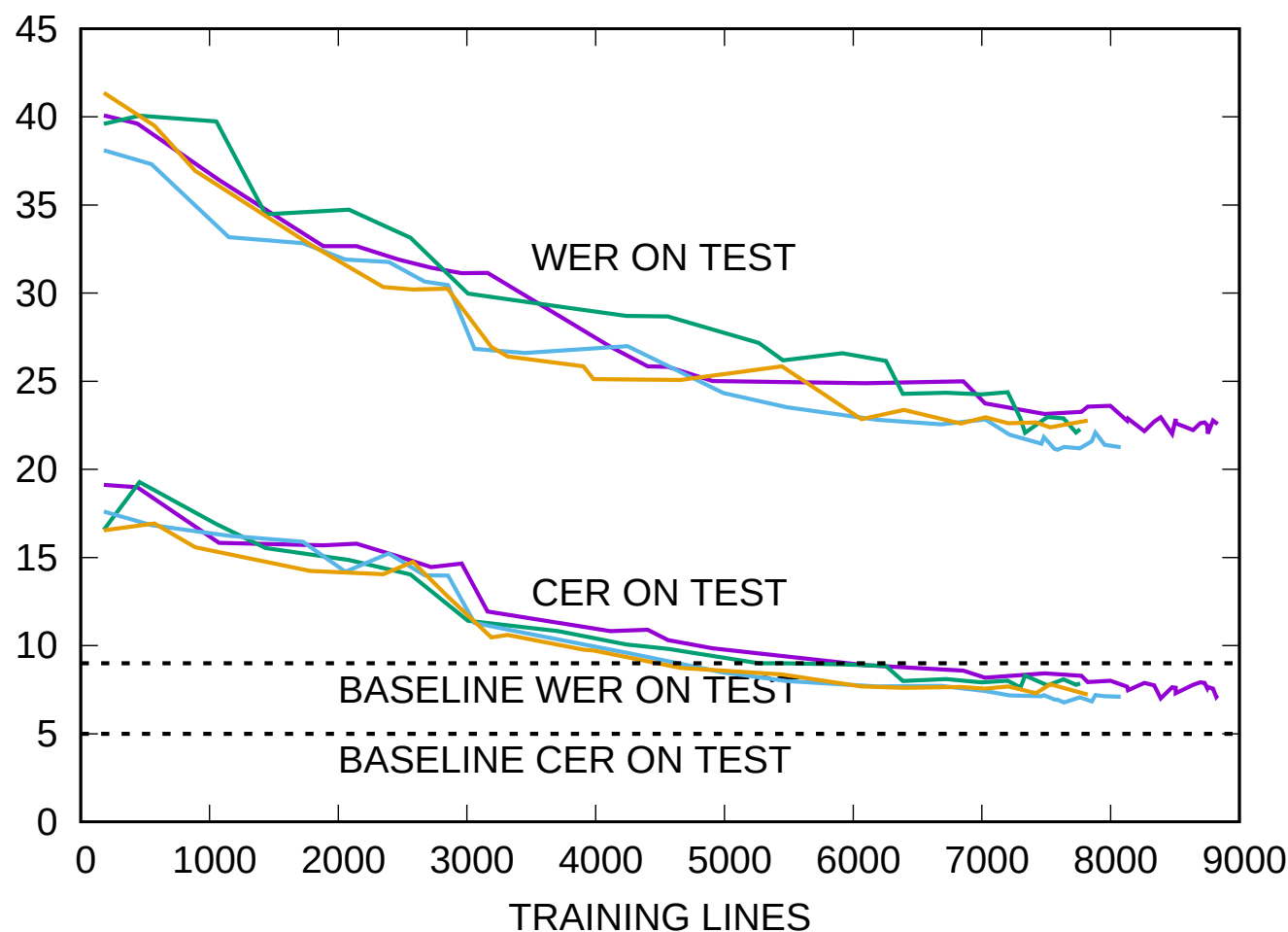
Scenario 2.1. Results with task dependent data for training the LM. 1 run



FINAL CER W/O LM: 8.0 FINAL CER WITH LM: 7.0
FINAL WER W/O LM: 🕒 FINAL WER WITH LM: 🕒

Semi-supervised learning

Scenario 2.2. Results with task independent data for training the LM. 4 runs



FINAL CER W/O LM: 8.3 FINAL CER WITH LM: 7.7
FINAL WER W/O LM: 25.5 FINAL WER WITH LM: 23.2

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2. Entropy measures for grammatical models
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6. Semi-supervised learning
7. **Active learning**

Active learning

Motivation: To select the data to be annotated in a intelligent way for preparing new GT. To reduce the human effort to prepare training data.

Active learning: a *learner* uses an uncertainty function for selection of training data that is subsequently annotated by an *expert* [Settles, 2008]

Uncertainty function for an unannotated training sample $\mathbf{x}$

$$\rho(\mathbf{x}) = - \sum_{\hat{\mathbf{y}}} P(\hat{\mathbf{y}}|\mathbf{x}, \Theta) \log P(\hat{\mathbf{y}}|\mathbf{x}, \Theta)$$

where $\hat{\mathbf{y}}$ ranges over all possible labels for input $\mathbf{x}$.

Active learning

Main idea

1. Train an initial model Θ with few training annotated samples $\mathcal{L}$
2. Compute informativeness of each sample in a set of unannotated samples $\mathcal{U}$ by means of the function $\rho()$ which is computed with current model Θ
3. A set $X_B^* \subseteq \mathcal{U}$ of B unannotated samples are selected according to the result of the ρ function
4. These samples X_B^* are then annotated by an expert
5. The set $\mathcal{L}$ is updated with the new annotated samples and they are removed from $\mathcal{U}$
6. Train a new model Θ with training annotated samples $\mathcal{L}$
7. Repeat steps 2 to 6 until some convergence criterion is satisfied

Active learning

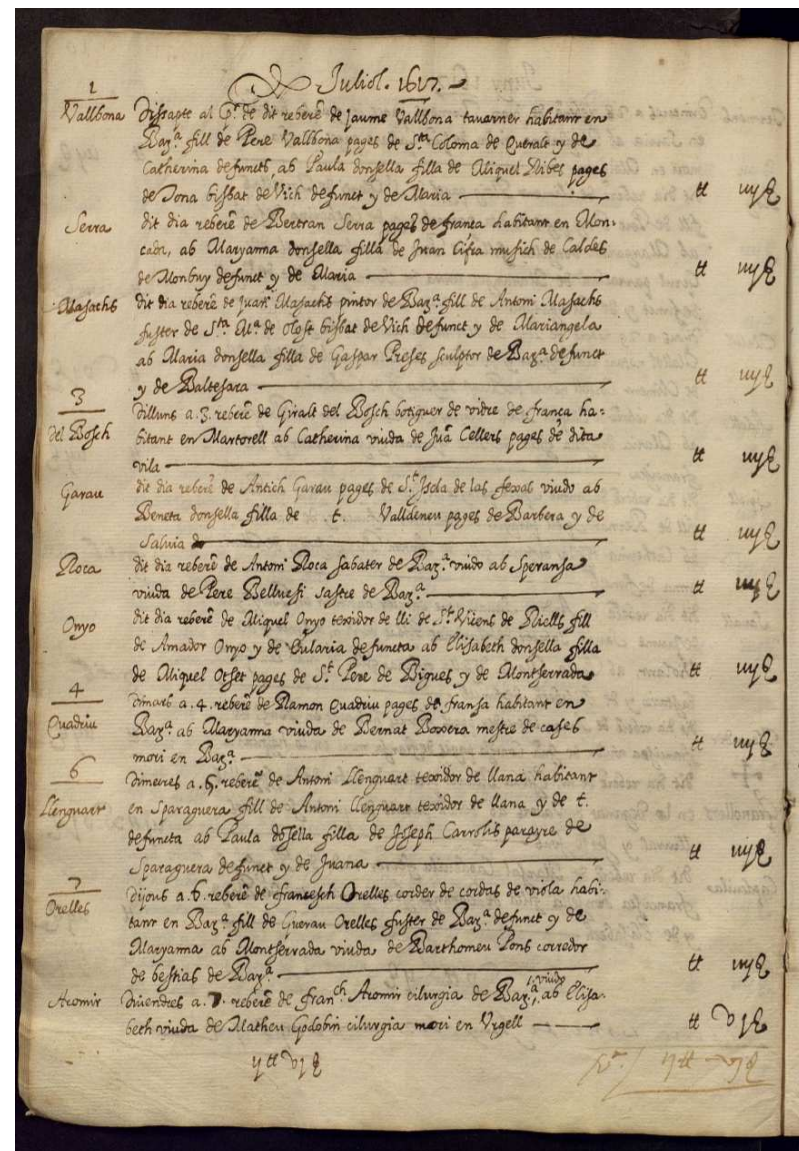
Experiments with the Esposalles dataset [Romero, 2013]

- Training: 4,620 lines, $\mathcal{L} = 150$, $\mathcal{U} = 4,470$
- Test: 827 lines

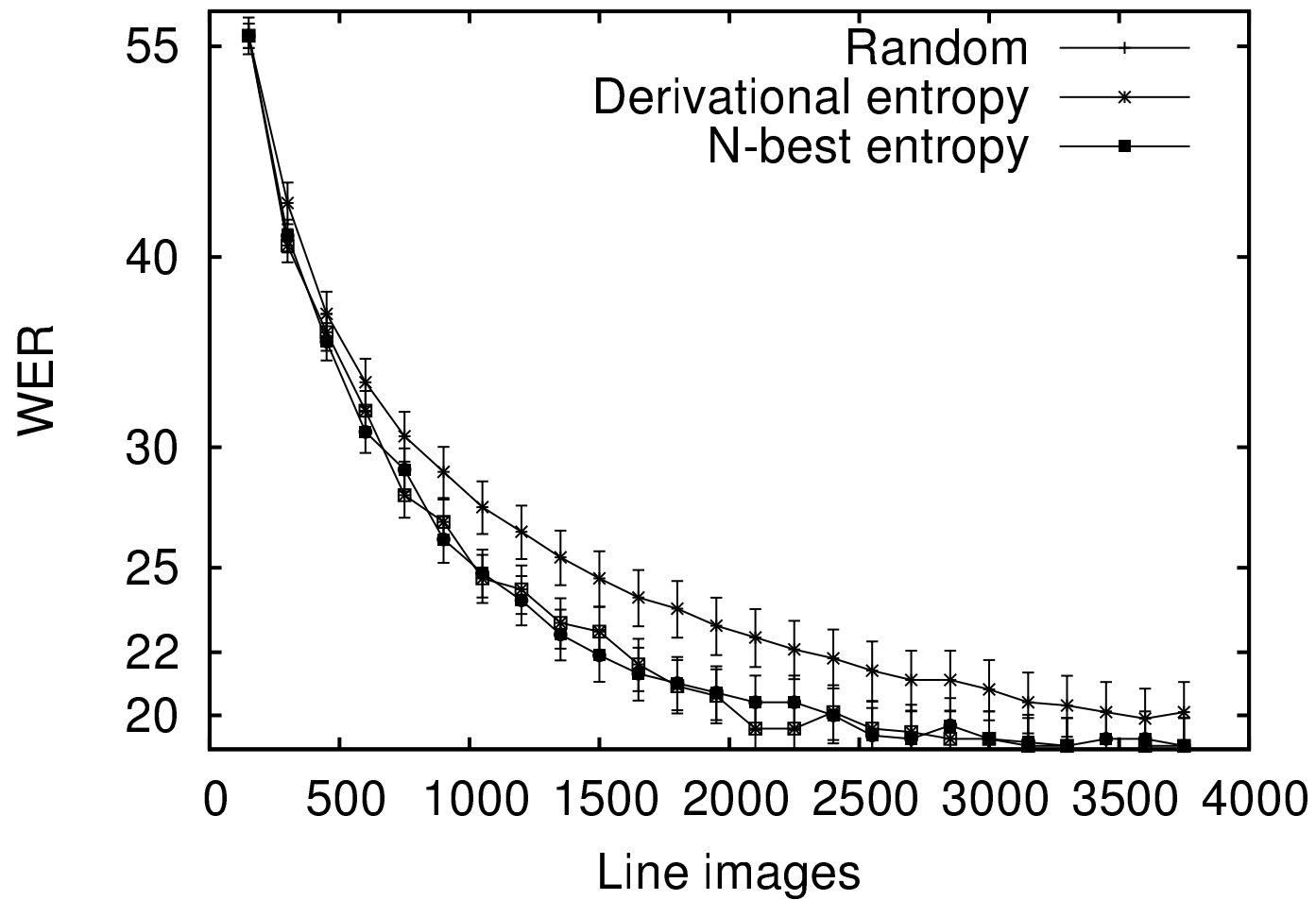
Optical models: Classical HMM

2-gram language model trained with $\mathcal{L}$ in each iteration

In each iteration $B = 150$ (about 5 pages) samples are selected



Active learning



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Appendix A. Entropy measures for grammatical models

$$\begin{aligned}
 & p(aba) \log p(aba) + p(abba) \log p(abba) + \dots \\
 &= .4 \cdot .3 \cdot .7 \log .4 \cdot .3 \cdot .7 + .4 \cdot .3^2 \cdot .7 \log .4 \cdot .3^2 \cdot .7 + .4 \cdot .3^3 \cdot .7 \log .4 \cdot .3^3 \cdot .7 + \dots \\
 &= .28 \cdot .3 \log .28 \cdot .3 + .28 \cdot .3^2 \log .28 \cdot .3^2 + .28 \cdot .3^3 \log .28 \cdot .3^3 + \dots \\
 &= .28 \cdot .3 (\log .28 + \log .3) + .28 \cdot .3^2 (\log .28 + \log .3^2) + .28 \cdot .3^3 (\log .28 + \log .3^3) + \dots \\
 &= .28 \cdot .3 \log .28 + .28 \cdot .3^2 \log .28 + .28 \cdot .3^3 \log .28 + \dots + .28 \cdot .3 \log .3 + .28 \cdot .3^2 \log .3^2 + .28 \cdot .3^3 \log .3^3 + \dots \\
 &= .28 \log .28 \sum_{n \geq 1} .3^n + .28 \cdot .3 \log .3 + .28 \cdot .3^2 2 \log .3 + .28 \cdot .3^3 3 \log .3 + \dots \\
 &= .28 \log .28 \sum_{n \geq 1} .3^n + .28 \log .3 \sum_{n \geq 1} n \cdot .3^n \\
 &= .28 \log .28 \frac{1}{1 - .3} + .28 \log .3 \frac{1}{(1 - .3)^2} \\
 &= .28 \log .28 1.43 + .28 \log .3 2.04 = -1.78
 \end{aligned}$$